

Competing for Quality: Coordination Failure in Dairy Genetics Markets

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<https://victorefunes.github.io/files/job-market-paper.pdf>

Abstract

The United States supplies nearly half of the world's traded dairy bull genetics, and genomic selection—an information technology that lets breeders read an animal's genetic merit at birth—has more than doubled the pace of genetic improvement since 2008. We show that the same technology intensified a coordination failure in this market. Breed-wide genetic diversity is a common-pool resource that any firm can draw down through inbreeding, and genomic testing raised the returns to inbreeding most for the sire lines already popular before its introduction, strengthening breeders' incentive to line-breed rather than outcross. Using National Association of Animal Breeders data on Holstein bulls marketed from 2001 to 2020, we compare popular lines with propensity-score-matched controls of comparable genetic merit in a difference-in-differences design. Inbreeding in popular lines rose by 1.2-3.5 percentage points - roughly 50 percent - relative to controls after 2010, with no differential pre-trend, a result robust to alternative matching estimators. A conceptual model links this behavior to a socially excessive drawdown of genetic diversity that can slow the industry's productivity growth over the long run.

Keywords: genomic selection; inbreeding; coordination failure; common-pool resources; dairy cattle genetics; difference-in-differences

JEL codes: Q16, O33, D62, Q13, L13

1 Introduction

The US plays a crucial role in the global dairy sector as one of the world's largest suppliers of dairy bull genetics. In 2022, the United States alone exported \$295 million of bovine semen, representing 47% of the world's exports in value terms (Figure 1). For about six decades, milk production per cow has consistently increased in the US herd, and more than half of this growth is estimated to be attributable to improvements in cattle genetics (Council on Dairy Cattle Breeding, 2025). Thus, the efforts of US breeders have likely had substantial impacts on the global dairy industry.

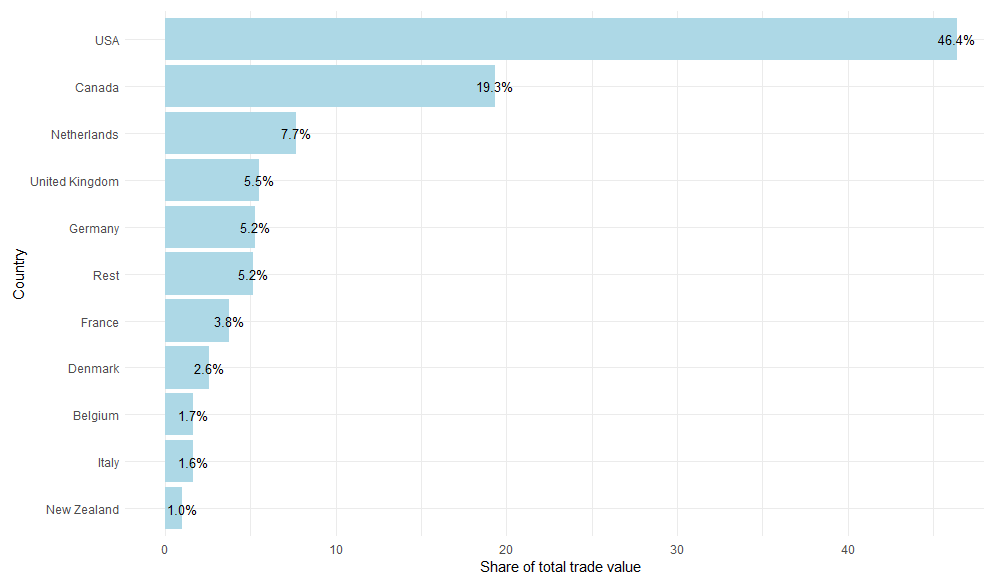


Figure 1: Share of trade Value of Bovine Semen exports by country (2019)

Source: UN Comtrade database

In the modern dairy industry, dairy breeders improve cow milk yield by identifying which bulls have the most productive female offspring and then selling this semen to dairy farmers looking to improve their genetics. The speed of this genetic improvement has previously been limited by biological factors. Determining whether a bull had good milk production ability took at least five years since the bull had to have offspring born from its use on dairy farms, and those offspring would need to take two years to grow to production age.

However, these limits became less binding after the advent of a technological innovation: genomic testing. Introduced in 2008, this technological innovation allows breeders to link each

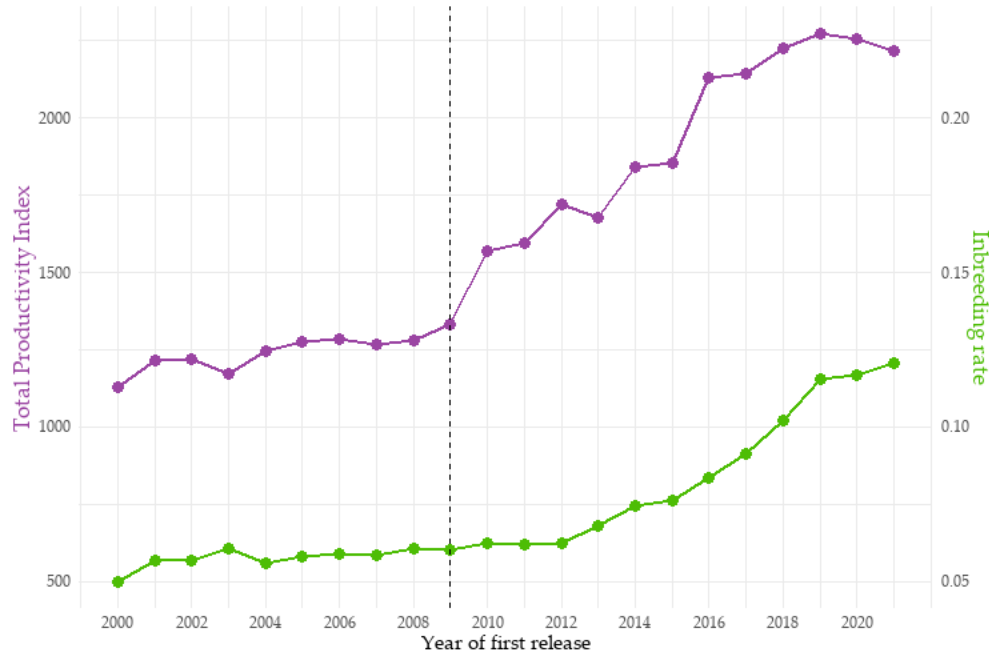


Figure 2: Dairy Bull Productivity versus Inbreeding Rate

Source: National Association of Animal Breeders (TPI)
and Canadian Dairy Network (Inbreeding rate)

animal's genes to productive outcomes such as milk yield and fat yield. The practical upshot is that determining a dairy bull's quality can now be done at birth rather than in 5 years. There are two important, documented impacts of genomic testing on the dairy bull market. First, the pace of genetic improvement in dairy cattle has more than doubled because breeders can identify productive bulls at much younger ages and breed new generations more quickly. Second, dairy companies have more than doubled the number of bulls for sale on the market since young bulls produce less semen than older animals; therefore, more animals are required to meet revenue goals.

The introduction of genomic testing has coincided with a less desirable outcome: an increase in inbreeding, the degree to which a dairy bull is related to the rest of the dairy cattle population. We focus our analysis on Holstein cattle, the dominant breed in the US dairy industry, representing roughly 90% of the US dairy herd and approximately 86% of bulls marketed by NAAB members during our sample period. Figure 2 shows the average inbreeding rate of Holstein dairy bulls on the market alongside the total productivity index (TPI), a measure of overall genetic performance

value¹.

Inbreeding is an inherent feature of pure-bred selection programs: concentrating genetic material from high-performing ancestors is a deliberate strategy to fix beneficial traits related to production, health, and fertility. However, excessive inbreeding, beyond what is individually or socially optimal, depletes genetic diversity and increases the risk of recessive disease alleles becoming expressed in the population. The welfare concern in this paper is therefore not inbreeding per se, but rather whether genomic testing has shifted breeders toward a market equilibrium with more inbreeding than is socially desirable.

Excessive inbreeding is harmful to the industry because it slows genetic improvement and can lead to costs down the road that seriously harm the dairy industry. In one case, breeding excessively from a famous dairy bull “Pawnee Farm Arlinda Chief” led to more than 500,000 spontaneous abortions and nearly 420 million dollars in losses because of an undesirable, recessive trait that was passed to his offspring (Adams et al., 2016). Given the US’s role in the global cattle genetics market, reducing the diversity of the dairy cattle gene pool could damage livestock markets worldwide.

In this paper, we examine whether the introduction of genomic testing increased inbreeding by increasing the returns to inbreeding relative to outcrossing for bulls who were well known in the industry. Genomic testing significantly increased the returns to inbreeding on these lines because, being the most widely used, the information gain from sequencing these lines was highest. Genomic selection also made it easier to detect productive genes that they want future offspring to inherit through inbreeding. Since well-known bulls are likely to be high-quality, we construct a set of counterfactual bulls comparable to popular genetic families in terms of productivity using propensity score matching. Using a difference-in-differences approach, we then examine inbreeding rates in popular families relative to our counterfactual families before and after the introduction of genomic testing in 2010. While the counterfactual of an industry without genomic testing is not observable, we focus on identifying the increase in inbreeding relative to bull lines where genomic testing would increase the returns to inbreeding.

We find that popular dairy bull lines became more inbred after the introduction of genomic

¹TPI is an economic value-weighted average of a set of m genetic traits: $H = \sum_{j=1}^m p_j g_j$, where p_j and g_j are the economic values and breeding values of trait j (Melton et al., 1979).

testing than in our counterfactual. The inbreeding rate of the most popular lines increased by 1.2-3.5 percentage points relative to our control group after 2010, representing about a 50% increase over pre-2010 rates. We also find no statistically significant difference in inbreeding rates across our groups before genomic testing was introduced. Using cost estimates from the Council on Dairy Cattle Breeding (CDCB), a back-of-the-envelope calculation implies that increased inbreeding in popular lines cost on the order of USD 1–3 billion over 2012–2019, depending on the share of the national herd descended from those lines (Appendix A).

This article fills a critical gap in the literature on livestock markets and the behavior of animal breeders. Few economics papers analyze the strategic actions of animal breeders, and even fewer analyze how their actions may impose costs on the industry. Early modeling of the animal breeder's decision dates back to studies such as Ladd and Gibson (1978), Kerr (1984), and Melton et al. (1994), which use hedonic regression and linear programming to determine the implicit market value of different livestock traits. Richards and Jeffrey (1996) and Schroeder et al. (1992) apply this method to dairy bulls to determine the economic value of each dairy bull trait implied by the results of a hedonic regression. Thus far, the economics literature has not recognized that animal breeders' decisions may lead to worse outcomes for the entire sector.

However, a related literature in agricultural economics has explored similar coordination problems among farmers controlling pests or disease. In the short run, farmers are incentivized to forgo the costs of controlling a future pest or disease because of a strong incentive to free-ride off of the efforts of others. This has been documented in the case of planting Bt-resistant crops (Brown, 2018, Epanchin-Niell and Wilen, 2015, Livingston et al., 2004, Wechsler and Smith, 2018) and managing citrus greening Lence and Singerman (2023).

Even in the absence of explicit externalities, individual profit-maximizing firms can find themselves in situations where product quality would be improved by coordination, even though coordination is not individually rational. Our paper is the first to explore such a coordination problem in the market for dairy cattle genetics. Breeders are incentivized in the short run to increase inbreeding by breeding cattle from popular lines, which may boost genetic improvement and capture more market share. However, inbreeding expends a type of resource, genetic diversity, that is needed to keep genetic improvement growing over time. Yet it is individually rational for firms to breed more than is socially optimal to maintain their market share by producing the

highest-quality bulls (Cole, 2019). Welfare loss occurs because maintaining a firm’s genetic diversity benefits the entire industry, as other firms can use it to restore their own diversity. However, an individual firm does not consider this benefit, leading to a slower rate of genetic improvement in the long run.

Before its introduction, it was theorized that genomic testing would **decrease** inbreeding by allowing breeders to more easily identify productive dairy bulls outside known genetic lines (Daetwyler et al., 2007). Instead, we find that the average inbreeding coefficient of dairy bulls increased along with dairy bull productivity. In our conceptual framework, we explore how genomic testing increases the returns to inbreeding and likely reduces genetic diversity across the industry. We provide evidence of this theory by comparing the inbreeding rates of groups of bulls that are similar in genetic traits but different in popularity before and after genomic testing. If genomic testing increased the discovery of less popular lines, we should expect no difference in the inbreeding rates of these groups, since animal breeders would have selected them solely on the basis of their genetic traits. Instead, we find that genetic lines already popular before genomic testing experienced an increase in their average inbreeding rate, thereby damaging the dairy sector.

If inbreeding continues to increase in the industry, even more damage might occur. Studies already estimate that the Holstein breed has an unsustainable level of inbreeding, having an “effective population” of between 43 and 66; it is commonly assumed that 50 is a critical level under which the breed will lose resilience (Makanjuola et al., 2020). Moreover, the lower the genetic diversity, the greater the risk that unknown recessive traits will cause significant damage to genetic lines (Cole and VanRaden, 2011, Lozada-Soto et al., 2024). At its worst, the population can reach the maximum possible increase in genetic trait values, leading to a stalling of genetic improvement. Future work in economics should examine how policy might be used to address market frictions and incentivize breeders to outcross, thereby mitigating the impacts of inbreeding.

The paper is structured as follows: Section 2 presents background and a conceptual model exploring how introducing genomic selection can lead to higher inbreeding rates in subsequent generations of dairy bulls. Section 3 explains the data sources and shows descriptive statistics. Section 4 outlines the identification strategy and methodological framework. In Section 5, we explain the results before doing a cost-benefit analysis in Section A. We conclude in Section 6 and discuss the implications for the future of the US dairy industry.

2 Conceptual Framework

2.1 The Coordination Problem in Dairy Breeding

To describe the dynamics of the dairy industry, we present a simplified theoretical model of the coordination problem inherent in dairy cow breeding. In the modern dairy industry, breeding and selecting new bulls for sale has become almost exclusively the job of dairy breeders. In contrast, dairy farmers are largely uninvolved in the breeding process and buy their genetics from these companies rather than producing or selling genetics themselves. Thus, we model the dairy breeder as a single breeding enterprise, not as a farm also engaged in milk production.

In our theoretical formulation, there is a market made up of N firms, and each firm i in time t produces quality q_{it} to earn a share of dairy industry revenue $R(\bar{Q})$, where $\bar{Q}$ is the average quality produced by all breeders. We assume that the dairy bull market has poor differentiation such that dairy farmers roughly treat every dairy bull as equivalent, and thus only pay attention to the amount of q_{it} the firm produces. This creates a zero-sum market environment, as a firm i that produces lower quality than its competitors (q_{-it}) will receive a lower share $s(q_{it}, q_{-it})$ of the market revenue $R(\bar{Q})$.

Firms have two options (a_{it}) to increase quality: inbreed (I) or outcross (O). The returns to these options depend either on the genetic diversity of the firm, d_{it} , or the genetic diversity of the other firms, d_{-it} . The actions in time t will determine the quality in time $t + 1$: $q_{it+1} = q_{it} + f(a_{it}, d_{it}, d_{-it})$, where f is:

$$f(a_{it}, d_{it}, d_{-it}) = \begin{cases} \gamma_I \cdot d_{it} & \text{if } a_{it} = I \\ \gamma_O \cdot d_{-it} & \text{if } a_{it} = O. \end{cases}$$

We assume that $\gamma_I > \gamma_O$ because inbreeding extracts more of the value from genetic diversity than outcrossing extracts from someone else's line. However, we also assume that choosing $a_{it} = I$ depletes d_{it} while $a_{it} = O$ replenishes d_{it} by adding genetic material from other lines. We assume the dynamics of d_{it} as:

$$d_{it+1} = \begin{cases} \delta \cdot d_{it} & \text{if } a_{it} = I \\ \min\{(1 + \rho)d_{it}, 1\} & \text{if } a_{it} = O \end{cases}$$

where $\delta \in (0, 1)$ is a depreciation rate and $\rho > 0$ is a replenishment rate. For simplicity, we assume that there is some upper limit to d_{it} that we can normalize to 1. The firm's tradeoff is whether to inbreed to increase q_{it+1} , and thus their revenue share next period $s(q_{it+1}, q_{-i,t+1})R(\bar{Q})$, at the expense of their stock d_{it} . Outcrossing would restore their genetic stock but might grant market share to their competitors because they will gain less q_{it} .

With the most basic cost function, one where a firm pays α_I to inbreed and α_O to outcross, the firm's maximization problem as a Bellman equation would be:

$$\begin{aligned} \max_{a \in \{I, O\}} \quad & V(a, q, d) = R(\bar{Q})s(q, q_{-i}) - C(a) + \beta V(q'(a), d'(a)) \\ \text{s.t.} \quad & C(a) = \alpha_I \{a = I\} + \alpha_O \{a = O\} \\ & \text{rules of motion for } d \text{ and } q \end{aligned}$$

where $\beta \in [0, 1)$ is the discount rate.

In many cases, the optimal action for a breeder would be to inbreed regardless of what the other firms do. Assuming only two players and two periods, this can be illustrated as a 2x2 game (assuming symmetric d and q):

		Breeder 2	
		Inbreed	Outcross
Breeder 1	Inbreed	$R(\bar{Q}_{II})/2, R(\bar{Q}_{II})/2$	$S_H R(\bar{Q}_{IO}), S_L R(\bar{Q}_{IO})$
	Outcross	$S_L R(\bar{Q}_{IO}), S_H R(\bar{Q}_{IO})$	$R(\bar{Q}_{OO})/2, R(\bar{Q}_{OO})/2$

where $S_L < 1/2 < S_H$ because $\gamma_I > \gamma_O$ and $\bar{Q}_{II}$ is the amount of quality produced from both players choosing I . A potential Nash equilibrium in this case is (I, I) , as inbreeding could be a profitable deviation when the opponent chooses to outcross. To be more exact, the condition for deviating to O when the opponent chooses I is:

$$\frac{\alpha_I - \alpha_O}{\beta} > R(\bar{Q}_{II})\frac{1}{2} - R(\bar{Q}_{OI})s_L.$$

Because $\gamma_I > \gamma_O$, $\bar{Q}_{II} > \bar{Q}_{OI}$ and, since $s_L < 1/2$, the right-hand side of the condition must be more than 0. Thus, the only time outcrossing would be profitable is when inbreeding is more costly than outcrossing, $\alpha_I > \alpha_O$, and the quality gain from inbreeding is sufficiently small. Since inbreeding reduces gains over time due to inbreeding depression, more realistically, we would expect the relative gain from inbreeding to decline at lower levels of d (i.e., a non-linear form for the gain function f)

The actions of both players will impact the aggregate diversity stock, $\bar{d}$. If each player inbred every period, we would see $\bar{d} \rightarrow 0$, leading to genetic improvement stagnating without any recovery. However, if inbreeding is more costly than outcrossing and the gains to inbreeding eventually slow, players would likely not let d drop below a threshold level, call it d^{ME} for “market equilibrium.”

Compare this to what we would assume a social planner would choose to maximize the discounted stream of welfare. Imagine that the social planner picks the level of genetic diversity d and that the welfare-maximizing value is d^* . We contend that $d^* > d^{ME}$, meaning that the market equilibrium is a socially inefficient level of genetic diversity. When a firm outcrosses, it benefits the firm and the entire industry by improving the benefits of outcrossing for all firms. The social planner takes into account both of these benefits, leading to a higher stock of genetic diversity in the long run. In contrast, an individual firm considers only its own benefit to its stock, leading to a lower level of genetic diversity. Assuming a welfare function W , the resulting welfare loss is $W(d^*) - W(d^{ME})$ since firms will use more diversity than is socially optimal to stay ahead of their competitors.

2.2 The Externality as a Common-Pool Resource Problem

The inefficiency in our model operates through a common-pool genetic resource rather than through prices. The Holstein breed is a single, closed population whose aggregate genetic diversity (summarized in our model by $\bar{d}$ and empirically by the breed’s effective population size (Makanjuola et al., 2020)) is reduced by inbreeding regardless of which firm performs it. This stock

has the two defining features of a common-pool resource. It is *rival*: coancestry fixed by one firm's line breeding reduces the number of effectively distinct gene copies circulating in the population, leaving less diversity for others to draw on when they outcross. And it is *non-excludable*: no firm can prevent others from breeding to the same widely-used sires, and the resulting rise in breed-wide relatedness, and in the probability that a recessive lethal such as that carried by *Pawnee Farm Arlinda Chief* (Adams et al., 2016) becomes fixed, is borne by the entire industry. This is the same structure studied in the management of shared biological stocks such as pest susceptibility and invasive species (Epanchin-Niell and Wilen, 2015), where any single agent's extraction depletes a resource available to all.

This mechanism is distinct from a pecuniary externality. A pecuniary externality is transmitted through market prices and, under standard conditions, requires no corrective intervention because the price change is itself efficient. The externality here instead acts directly on the biological stock $\bar{d}$: a firm that inbreeds lowers the diversity available for every other firm's future outcrossing, independent of any effect on semen prices. It is this uncompensated, quantity-based effect on a shared stock, not a price effect, that drives the wedge between the market equilibrium d^{ME} and the social optimum d^* .

The common-pool structure also distinguishes our explanation from one based simply on impatience. A natural alternative is that breeders are privately rational but heavily discount the future, so that any over-inbreeding reflects individually optimal behavior that requires no policy response. High discount rates do exist in this industry and would worsen the problem (Wuepper et al., 2023). They are not, however, its source. Even a firm with a socially efficient discount rate would inbreed beyond d^* , because a large share of the long-run cost of its choice (reduced diversity for competitors and future entrants, and heightened breed-wide exposure to fixed recessives) falls on others rather than on itself. In the model, this is exactly the statement that each firm optimizes over its own stock d_{it} while ignoring its contribution to $\bar{d}$. The inefficiency survives even at the socially correct discount rate, which is what marks it as a genuine externality rather than a preference.

2.3 The Effect of Genomic Testing

Genomic testing allows dairy breeders to determine the genetic traits of dairy bulls far more quickly than before. Before it was introduced, it took five years to determine a dairy bull's genetic traits. Once the bull reached sexual maturity around the age of one year, the bull's semen would be frozen and sent to several farms to produce offspring from the bull. After about 10 months, the offspring would be born and would take roughly another 2 years to reach production age. After a few years of production, the bull now has production data from its offspring that can be used to calculate its genetic traits, in our model q .

Genomic testing gathers the same information in a much shorter time by taking a DNA sample from the bull and correlating its genes with traits. In our model, genomic testing accelerated the rate of quality gain, that is, increased γ_I and γ_O . In the case that they both increase the same amount, there would be no change in d^{ME} since the relative benefits of inbreeding have not changed. However, genomic testing disproportionately benefits inbreeding by making it easier to identify beneficial genes. In particular, it makes it easier to identify which offspring inherited the genes, so isolating the beneficial gene is easier than before. Since outcrossing is about expanding the number of genes in the firm's genetic families rather than isolating a specific gene, it benefits far less from genomic testing. Moreover, this isolation works best when there are fewer genes to test for in the population, so genomic testing can more efficiently isolate a population that shares fewer genes (i.e., a population with lower genetic diversity).

In the model, this would increase γ_I relative to γ_O , leading to a new "genomic" market equilibrium level of genetic diversity, $d^{GE} < d^{ME}$. Since inbreeding is now more valuable, the new social planner optimum may also be lower: $d^{**} \leq d^*$. Whether genomic testing actually leads to a higher welfare loss depends on i) how much genomic testing decreases d^{GE} and d^{**} and ii) the relationship between W and d . If $d^* - d^{ME} < d^{**} - d^{GE}$, that is, the diversity gap increased, then genomic testing causes a net welfare loss if welfare is improved if $W'(d) > 0$ or diversity increases welfare. If $d^* - d^{ME} = d^{**} - d^{GE}$, the welfare impact of genomic testing depends on whether d is linear or concave in W . If d is linear in W , then $W(d^{**}) - W(d^{GE}) = W(d^*) - W(d^{ME})$, meaning the welfare loss is unchanged. However, if d is concave in W , then $W(d^{**}) - W(d^{GE}) > W(d^*) - W(d^{ME})$, since they are at lower values. Since inbreeding depression would cause the

benefits of inbreeding to erode over time, this is most consistent with d being non-linear in W .

The primary prediction of this model is that the introduction of genomic testing increases inbreeding, which may exacerbate welfare loss in the industry. The welfare loss arises from breeders excessively inbreeding to stay ahead of their competitors, which depletes genetic diversity and leads to slower aggregate quality growth over time. In this paper, our aim is to test one prediction of the model: that genomic testing increases inbreeding. Specifically, we hypothesize that after genomic testing, the lines that become the most inbred are those that were the most popular before genomic testing. Since most bulls would share DNA with these lines, testing their DNA would provide firms with the most information. Taking our model to data, we would then expect inbreeding to increase the most in the popular lines irrespective of quality, as genomic testing will have made inbreeding more beneficial on these lines than before. This would be consistent with the returns to inbreeding increasing relative to outcrossing.

3 Data Description

To understand the role of genomic testing in dairy breeder decision-making, we use data on dairy bull traits and lineages, including the inbreeding rate of each bull. Dairy bull traits in the US are calculated using data on dairy bull offspring producing milk on US dairy farms. The National Dairy Herd Improvement Program (NDHIP) collects, manages, and analyzes data from dairy farms that are members of Dairy Herd Improvement Associations to assess the bull’s contribution to the productivity of their offspring (Hutchins and Hueth, 2023). The raw data collected from farms are used to estimate a statistical model that predicts the genetic contribution of an animal as a sire (or dam) after controlling for environmental variables such as herd, age, pedigree, and season effects. The predicted values for each trait from this model are commonly called the **predicted transmitting ability** (PTA) of a trait.

The primary data for our study are from the National Association of Animal Breeders (NAAB), a trade organization representing all major livestock genetics companies that sell dairy bulls in the US. As noted in the introduction, we restrict our sample to Holstein bulls, the dominant breed in the US dairy herd, for whom the NAAB data is most comprehensive. Dairy bull traits for all bulls sold by members are posted three times a year at the same time that each bull’s predicted

genetic traits are calculated and posted publicly by the Council on Dairy Cattle Breeding (formerly by the USDA) (Hutchins and Hueth, 2023). Our data comes from the NAAB’s published lists from 2001 to 2020 and represents over 15,400 dairy bulls, both foreign and domestic, sold by its members during this period. The Association represents 95% of the livestock genetics companies and cooperatives in the US and Canada,² making these data broadly representative of what is sold to US dairy farmers.

The NAAB lists also contain lineage information, which we use to track a bull’s descendants. Every animal born in the USA or Canada is assigned a unique ID that includes information about its breed and country of birth. The NAAB data include these IDs for the bull and its two parents, allowing us to build a genetic network of relationships.

$$F_X = \sum_{ca=1}^K \left(\frac{1}{2}\right)^{n_1+n_2+1} (1 + F_{ca}) \quad (1)$$

The **inbreeding coefficient**, F_X (Wright, 1922), defined in equation 1, is calculated for each bull at birth based on the pedigree records of its parents and ancestors. Where ca denotes the common ancestor, n_1 and n_2 are the numbers of generations separating the common ancestor c_a from the sire and the dam, respectively, and F_{ca} is the inbreeding of the common ancestor. The inbreeding coefficient F_X is the animal-level empirical counterpart to the depletion of d_{it} in our conceptual framework: a bull produced by line breeding (action I) inherits a larger share of identical-by-descent alleles, raising F_X , so a rising average F_X within a line is a direct signal of the decline in that line’s diversity stock d that the model predicts.

In practice, this calculation is performed by the Council on Dairy Cattle Breeding (CDCB) using pedigree data collected through the National Dairy Herd Improvement Program (NDHIP), and is reported each time a bull’s genetic evaluations are published, three times per year. The inbreeding coefficient used in our empirical analysis is drawn from the NAAB dataset, which reports the CDCB-calculated inbreeding coefficient for each bull at the time of its first release. This figure is calculated using the full national pedigree database and is therefore representative of the entire US Holstein population served by NAAB members, rather than a regional or membership-restricted subsample.

²<https://www.naab-css.org/history>

Figure 2, by contrast, uses the population-average inbreeding rate reported by the Canadian Dairy Network (CDN) for illustrative purposes only, to show the industry-wide trend in inbreeding alongside productivity growth. The CDN reports a population-average inbreeding coefficient calculated using the same Wright (1922) methodology applied to the Canadian Holstein pedigree, and trends in Canadian and US Holstein inbreeding rates are closely comparable given the high degree of genetic exchange between the two countries' breeding programs. This figure is not used in any of our empirical estimations, which rely exclusively on the NAAB-sourced inbreeding coefficients.

The genomic testing program began in 2008, and the first genomically tested bulls had their evaluations published in January 2009. García-Ruiz et al. (2016) finds that, after 2010, the generation interval decreased significantly by about four years. Using the list of bulls that are actively sold by members of the National Association of Animal Breeders, we can see in Figure 3 that the new genomically proven bulls, first listed in 2010, have an average age of a little less than two years old when they are first released, versus around five years for daughter-proven bulls.

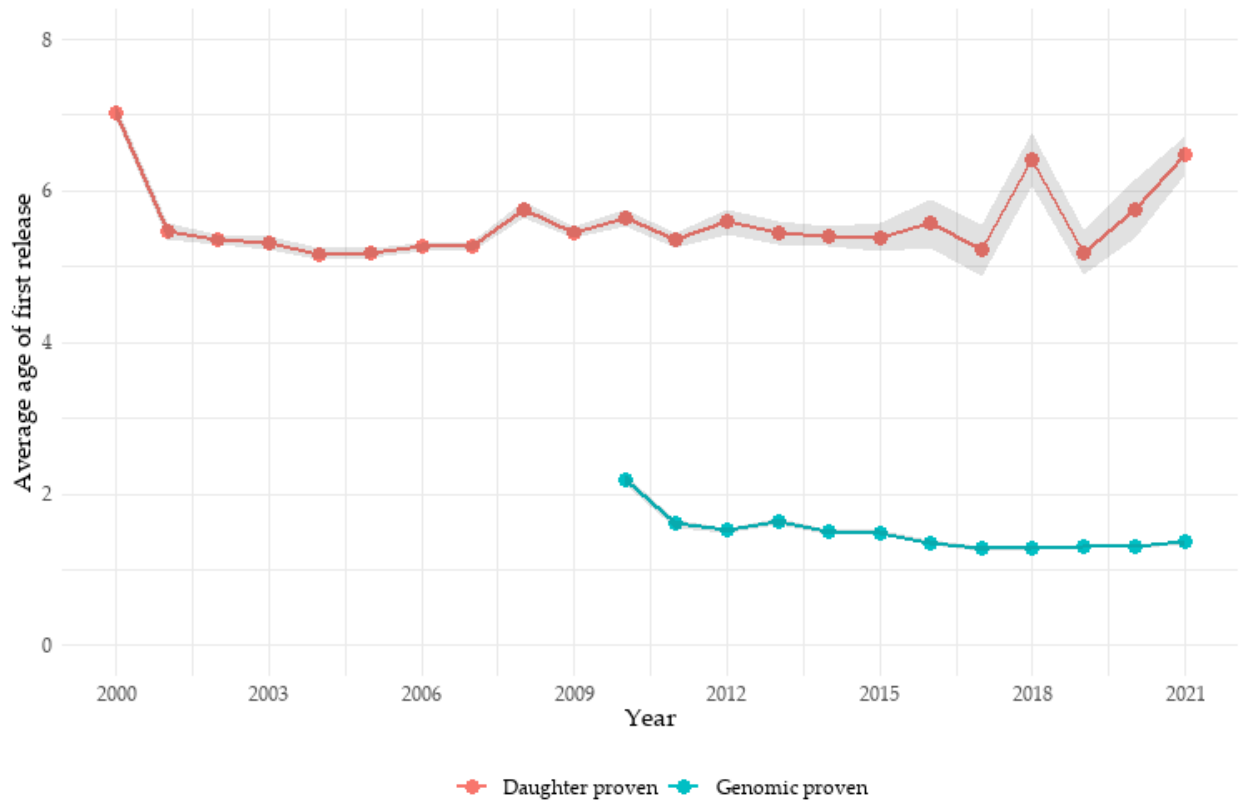


Figure 3: Average age at first proof for Holstein bulls
Source: NAAB

In addition to a significant increase in genetic milk yields, the number of bulls marketed annually has also increased substantially (Hutchins and Funes-Leal, 2025).

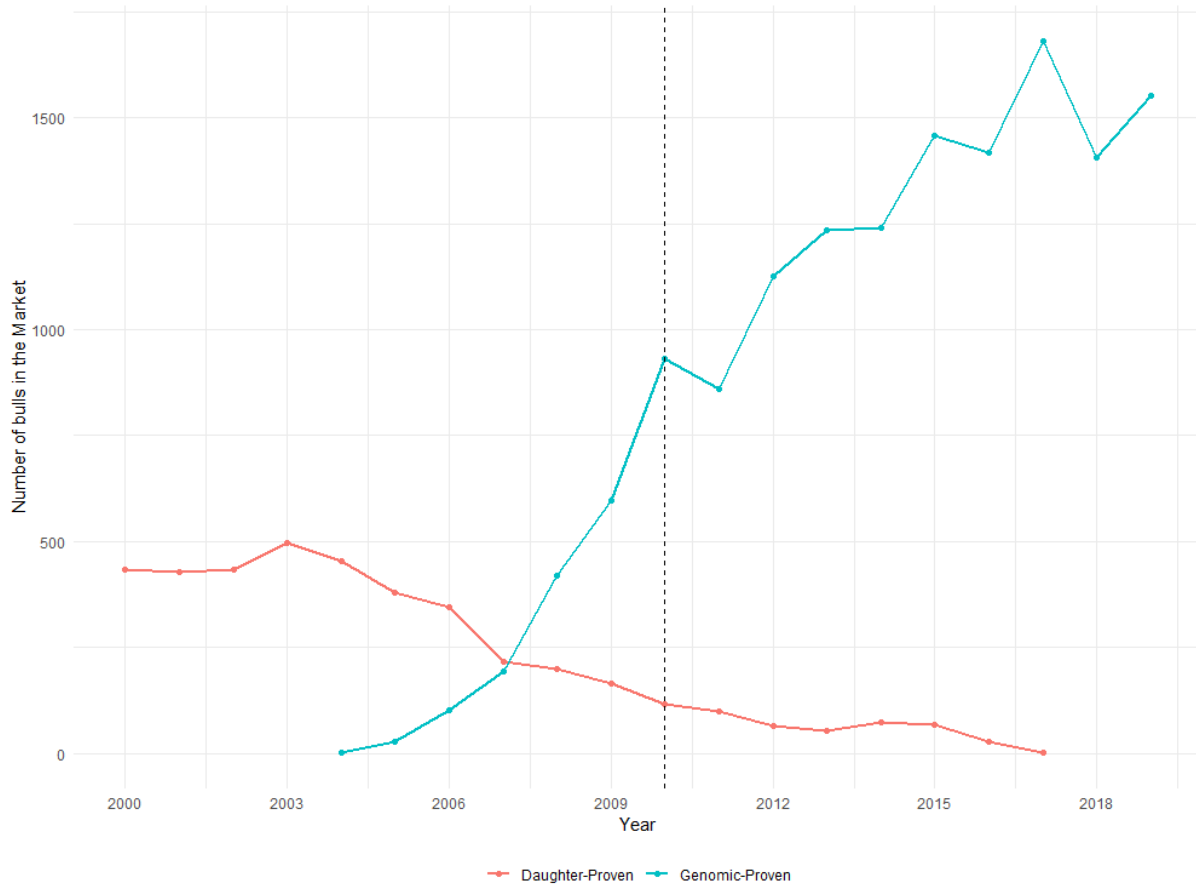


Figure 4: Total number of new bulls released in the market
Source: NAAB

Figure 4 shows that the number of bulls listed as actively sold in the industry tripled starting in 2010. Such an increment is motivated by the ability to detect bulls with high genomic merit as soon as they reach sexual maturity, rather than waiting for daughter records. Because these young bulls produce less semen than mature animals, more of them must be marketed to meet the same revenue, which helps explain the roughly threefold rise in listed bulls (Figure 4). As seen in the plot, genomically-tested bulls have become the dominant type of testing in the dairy genetics industry.

Figure 5 shows the percentage of bulls that are tested by birth year.

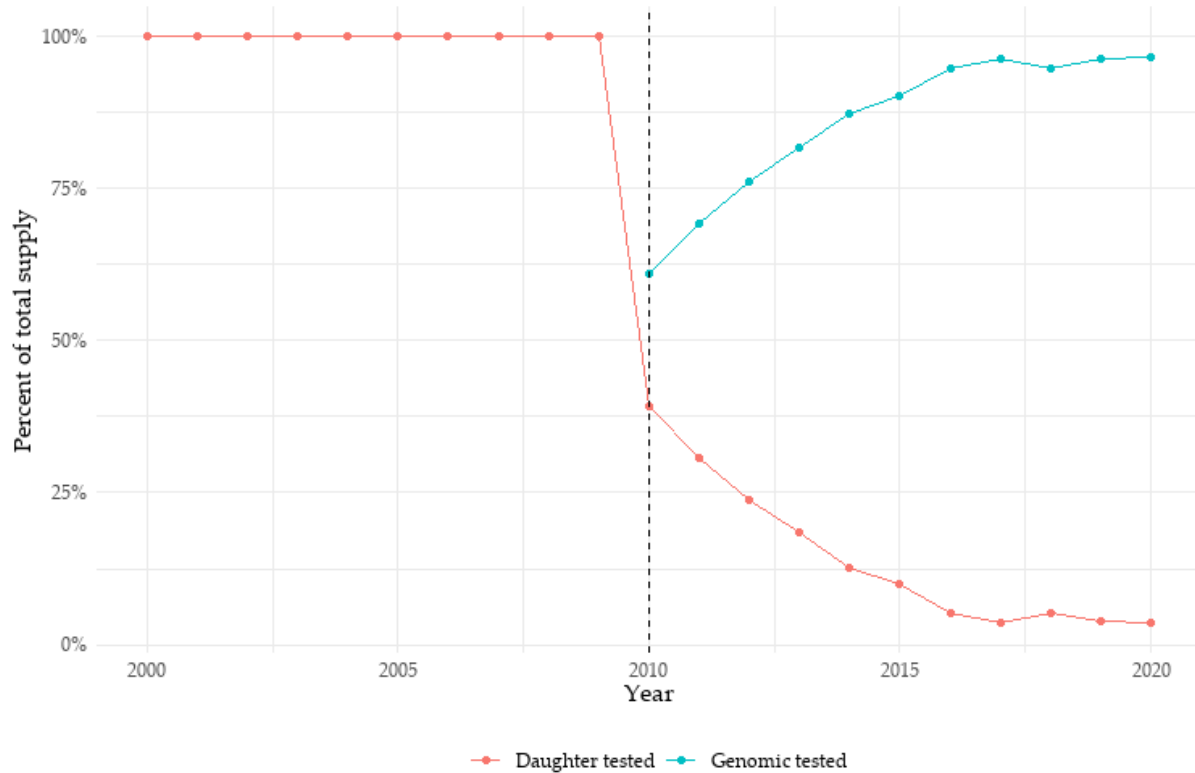


Figure 5: Share of each status of bull by year

Figure 5 depicts that, by 2017, genomically-tested bulls accounted for approximately 95% of all new Holstein bulls released to market.

4 Empirical Framework

The goal of this analysis is to determine whether genomic testing increased inbreeding in cattle by increasing the returns to inbreeding among popular genetic lines. Given the high percentage of genomic adoption in the data, it is unlikely we can use ungenotyped bulls as a control group. In an empirical approach, we focus on comparing lines with similar genetic quality but differing in pre-existing popularity, measured by the number of sons before genomic testing. We hypothesize that genomic testing makes inbreeding more valuable on these genetic lines because it is more accurate and there are fewer genes to test. This would suggest that genomic testing has caused breeders to deplete genetic diversity even more than they would have before.

To isolate the role of the existing popularity of genetics, we construct a counterfactual sample

with the same genetic traits but from a less popular line. Specifically, we use propensity score matching (Smith and Todd, 2005) to define a control group that resembles the treatment in all observable characteristics, in this case, the genetic traits. If genomic testing led to a race for quality bulls regardless of prior popularity, we should see the inbreeding rate of these lines increase by the same amount as the popular lines. If, instead, genomic testing accelerated line breeding for the popular lines, we should see a significant difference in the inbreeding rates after 2010.

We define a **line** as the set of direct (male) descendants from a particular bull. We also define a “popular” bull as a bull whose number of sons is in the 95th percentile of the distribution of offspring. To construct the line, we start by considering sires born between 1998 and 2004, as the “founders” of each line. Our definition depends critically on when the founders of such lines were born; if we choose too early a date, lines are too difficult to distinguish since all bulls are related to some degree. After identifying the sires on which to build lines, we recursively build their pedigrees by matching each bull to its sire (father) until we reach the latest animal in the sample. Figure 16 in the appendix summarizes the sample selection mechanism; the treatment and control groups comprise the descendants of two groups: popular bulls (treatment) and matched bulls (control).

Breeders are under high competitive pressure to supply bulls with the highest PTAs. However, the best way to do so is by linebreeding a family of bulls so that the relative distance to an ancestor remains constant across generations. Steyn et al. (2022) and Steyn et al. (2023) show that younger sires capture a more significant share of present-day population genetic variation; hence, genotypes of older sires can be discarded from (genomic) evaluations. As generations begin to accumulate, the contribution of a particular sire decreases (on average) exponentially, so we can disregard the impact of distant ancestors.

The first step is to match the founders of such lines with other sires that did not father as many sons but had comparable genetic traits. Second, we find all descendants from both groups and compare the evolution of their inbreeding rates through time. The idea behind this procedure is that when those bulls were in the market, any reasonably similar sire could have been chosen but was not, and thus, can be used to construct a counterfactual line. To do so, we use a propensity score matching estimator, let $p(X) = \Pr(T_j = 1|X)$ where T is a dummy equal to 1 if sire j 's offspring is on the 95th percentile of the distribution and X_j be a vector comprised of the following:

1. Predicted Transmitting Ability for production traits (milk, fat, and protein), health traits (mastitis, metritis, productive life), and reproductive traits (daughter pregnancy rate, early first calving, gestation length).
2. Dummies for the absence of certain recessive genetic disorders (Holstein Foundation, 2021) such as *brachyspina*³, complex vertebral malformation, cholesterol deficiency, and mule foot.
3. Dummies for the presence of genes associated with specific qualitative traits (polled, or red coat color).
4. Dummies for the presence of haplotypes (specific sequences of DNA at different locations in the chromosome) that affect fertility (Holstein Foundation, 2018) when present in homozygous form (likely to be expressed when two carrier animals are mated).

We then estimate the propensity score (Cameron and Trivedi, 2005, p. 873), the probability of being a popular bull given the level of PTAs and the presence of the entire set of variables listed above. The validity of our method relies on a series of assumptions (Heckman et al., 1997):

Assumption 1 (Stable Unit Treatment Value Assumption): $(inb^{treatment}, inb^{control}) \perp\!\!\!\perp T/X$.

SUTVA requires two conditions to hold (Imbens and Rubin, 2015). First, the potential outcomes of one unit should not be affected by the treatment assigned to other units (no interference). Second, there should be only one version of each treatment level (no hidden versions of treatment).

The second condition is plausibly satisfied in our setting: genomic testing was introduced as a single, industry-wide technological change in 2008, and all firms faced the same testing technology. There are no meaningfully different "versions" of genomic testing that would affect some lines differently from others through the treatment itself.

The first condition, however, warrants more careful discussion. Our conceptual framework explicitly models a coordination failure in which one firm's inbreeding decisions affect the genetic diversity available to other firms for outcrossing. In this sense, SUTVA may be partially violated by construction: if popular lines become more inbred after genomic testing, this mechanically reduces the diversity available to other lines, potentially affecting their inbreeding rates as well.

³A lethal genetic disease that causes spontaneous abortions.

A strict reading of the no-interference assumption would therefore be difficult to defend in this setting.

We address this concern in two ways. First, our identification strategy focuses on the differential effect of genomic testing on popular versus non-popular lines, rather than on the absolute level of inbreeding in either group. Cross-line spillovers would need to affect popular and non-popular lines asymmetrically and in a way unrelated to their pre-existing popularity to bias our estimates. Second, the matching procedure ensures that treatment and control founders are comparable in all observable genetic traits prior to genomic testing, so any spillovers that affect both groups symmetrically will differ out in our estimator. We therefore interpret our estimates as the differential causal effect of genomic testing on inbreeding in popular lines relative to comparable non-popular lines, rather than as an absolute treatment effect on inbreeding for the industry as a whole.

Figure 6 shows the extent to which the inbreeding rate distribution of our matched sample resembles that of the treatment group. Unlike unmatched sires, their inbreeding distributions have similar means and variances conditional on the set of genetic traits. In a similar fashion, Figure 15 shows the standardized mean differences of all variables used to estimate the propensity score. Balance is achieved for several variables, notably for all haplotype dummies and several PTAs.

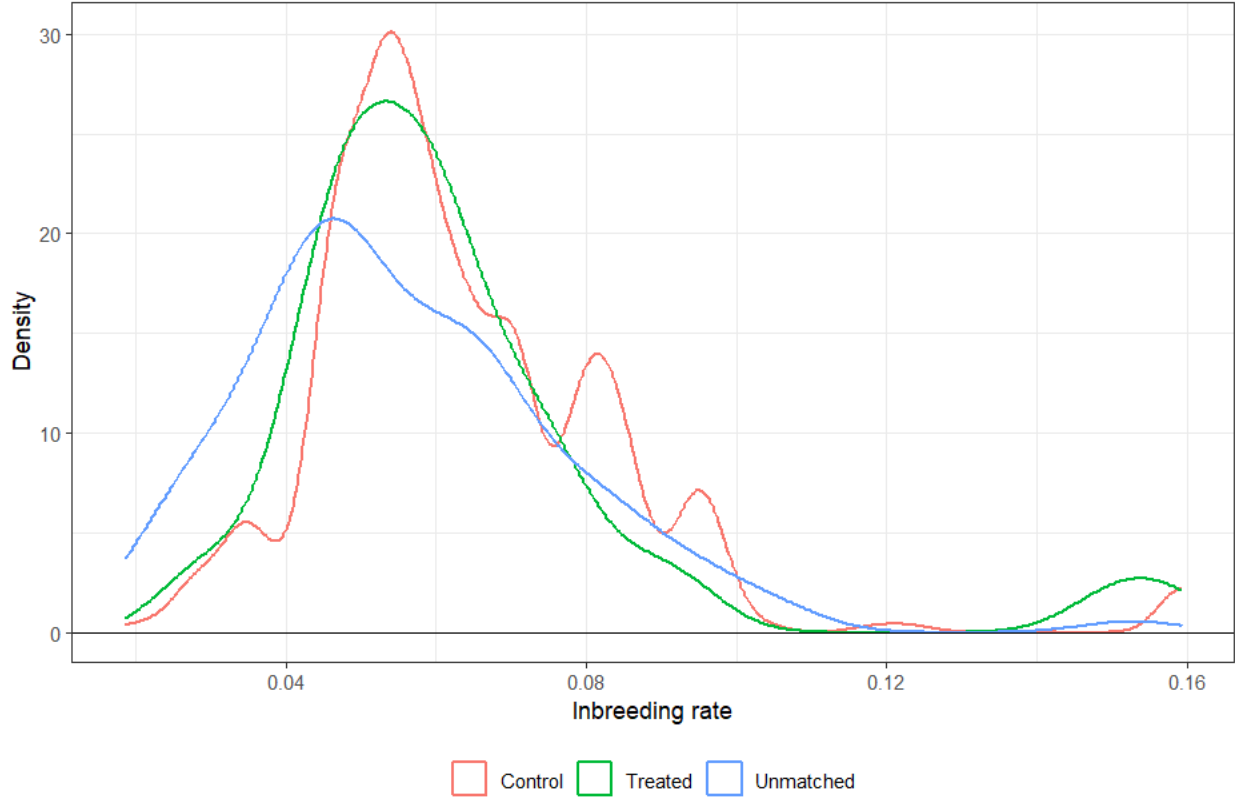


Figure 6: Distribution of Inbreeding rates across pre-treatment groups

Assumption 2: $0 < \Pr(T_j = 1|\mathbf{X}) < 1$ for all $\mathbf{X}$

This assumption implies that we can define a propensity score for every value in the treatment group. Assumptions 1 and 2 together are the “strong ignorability” assumption in Rosenbaum and Rubin (1983), and they are necessary in conjunction to identify causal parameters. The set of matching variables above exhausts the entire set of possible selection criteria for breeders; no unobserved variable can influence mating decisions. Figure 7 shows the degree of overlap between popular bulls and alternative sires; since there is a considerable difference in the size of the groups, popular bulls make up only a tiny fraction of bulls; however, the overlap is good in the range 0.05 – 0.50, but there exist a few popular bulls that had few close matches in the sample.

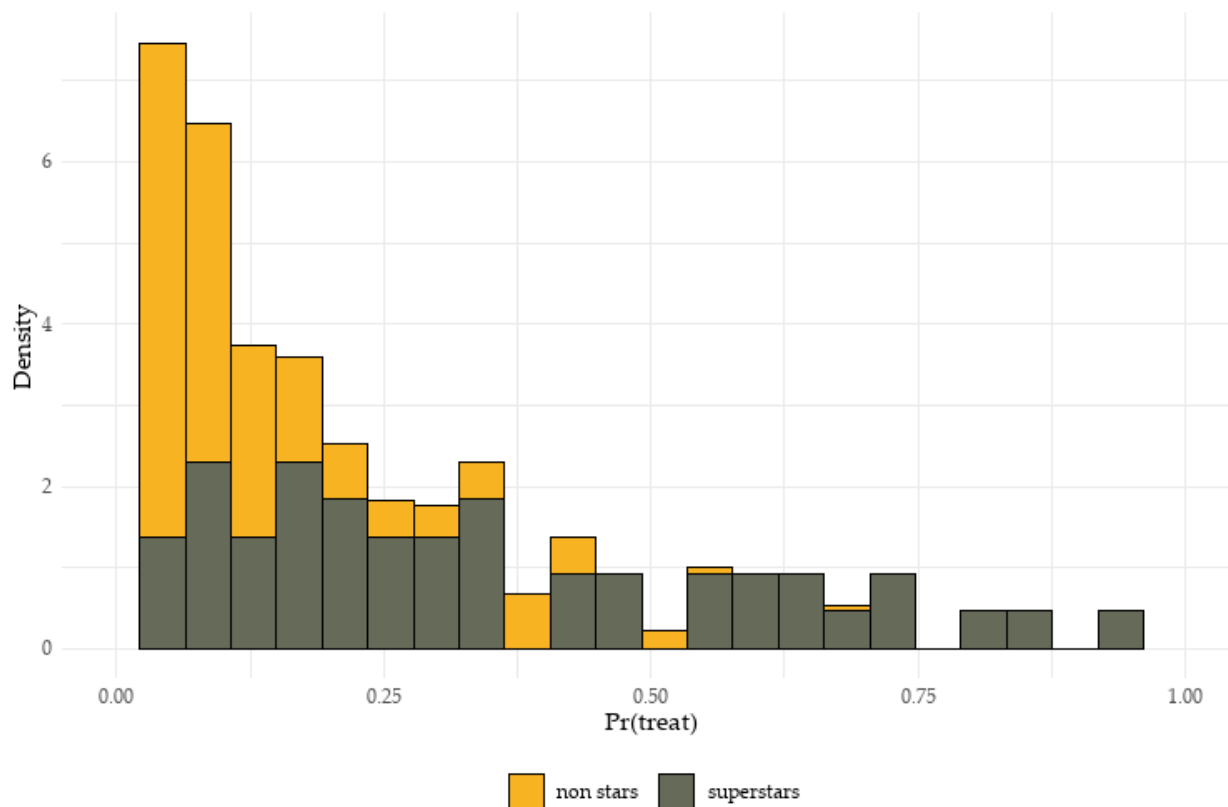


Figure 7: Propensity score by class of bull

A central part of our argument is that treatment and control groups should have similar inbreeding levels before genomic selection. To test that assumption, we calculate the difference between the averages of the treatment and control groups for inbreeding rates and selected traits in Table 1. There is no statistically significant difference in most traits between the treatment and control groups, so our matching method ensures that the only difference between the two groups is that one bull has been used more often by dairy farmers. Each year, these bulls will have a new set of descendants being born, each with its own inbreeding rate and genetic traits; if our hypothesis holds, the inbreeding rates of the descendants of popular bulls should be significantly higher than those of non-popular descendants.⁴

⁴In Appendix C, we discuss three more assumptions that must hold for our estimator to be valid, including parallel trends, no anticipatory effects, and asymptotic convergence.

Table 1: Balance table for outcome and matched covariates

Variable	Control mean (n = 224)	Treatment mean (n = 43)	t.statistic	p.value	df
Inbreeding rate	2.968	2.904	0.426	0.672	58.147
Milk	0.059	0.059	-0.036	0.971	59.956
Fat	-0.122	-0.170	0.346	0.730	76.548
Protein	-0.851	-1.009	0.416	0.679	61.059
Somatic cell score	-0.151	-0.037	-0.675	0.502	66.764
Productive life	-0.619	-0.677	0.173	0.864	59.829
Daughter preg. rate	-1.224	-0.489	-1.336	0.186	70.101
Heifer conc. rate	-11.953	-5.553	-1.807	0.075*	71.841
Cow conc. rate	-0.420	-0.209	-1.278	0.206	65.480
Livability	-0.393	-0.177	-1.816	0.074*	70.905
Type	-0.827	-0.551	-1.721	0.090*	69.510
Gest. length	0.342	-0.055	1.719	0.090*	64.087
Heifer liv.	-0.569	-0.579	0.032	0.975	58.933
Early first calving	0.101	0.232	-1.549	0.127	58.339
Mastitis	-0.888	-0.764	-0.288	0.774	58.621
Metritis	-0.840	-0.891	0.202	0.840	62.094
Strength	-0.589	-0.581	-0.047	0.963	57.734
Rear legs rear view	-191.265	-24.936	-1.581	0.119	67.713
Rear legs side view	-0.027	-0.013	-0.495	0.623	60.964
Foot leg score	-1.348	-1.123	-0.666	0.508	59.099
Teat rear place	-8.284	-3.255	-1.859	0.067	68.421
Rump angle	-0.360	-0.136	-1.447	0.153	64.334
Thurl width	-0.010	0.050	-0.342	0.734	63.537
Foot angle	-0.767	-0.417	-2.044	0.045**	68.106
Fore udder	-0.743	-0.432	-1.735	0.087*	66.269
Rear udder height	0.142	0.193	-0.254	0.800	60.330
Rear udder width	3.044	2.995	0.690	0.493	52.692
Udder cleft	-0.386	-0.349	-0.235	0.815	77.188
Udder depth	-0.206	-0.267	0.409	0.684	67.108
Teat fron place	-0.343	-0.195	-1.006	0.318	66.497
Teat length	0.197	0.159	0.195	0.846	59.002
Milk fever	-0.312	-0.140	-0.948	0.347	62.811
Stature	-0.255	-0.233	-0.138	0.890	72.664
Dairy form	-0.573	-0.403	-1.442	0.153	73.087
Body depth	-0.401	-0.193	-1.127	0.264	61.777
Sire calv. ease	-0.738	-0.589	-0.989	0.326	73.000
Daughter calv. ease	2.427	2.287	1.188	0.239	63.963

Notes:

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

We observed the average inbreeding rates of the treatment and matched control groups before and after the introduction of genomic selection. Let $j = 1, \dots, J$ be the set of lines, defined as the set of direct descendants from a single male ancestor. Each line is comprised of $i = 1, \dots, n_j$ bulls born in year t and belongs to firm $k = 1, \dots, K$. If we divide the 2005-2017 period into two groups, pre-intervention (2005-2009) and post-intervention (2010-2019), denoted as 1 and 2 and let $treat_j$ denotes whether a bull from line j belongs to treatment ($treat_j = 1$) or control ($treat_j = 0$) groups, then:

$$inb_{ijkt} = treat_j \times inb_{ijkt}^{treat} + (1 - treat_j) \times inb_{ijkt}^{control}$$

Where inb_{ijkt} is the inbreeding rate of bull i from line j , that was marketed by firm k and was born in year t . To measure the effects of genomic selection on inbreeding, we estimate the differences-in-differences model:

$$inb_{ijkt} = treat_j + post_t + (treat_j \times post_t)\beta + \varepsilon_{ijkt} \quad (\text{Model 1})$$

$$inb_{ijkt} = treat_j + treat_j \times \sum_{\substack{t=2005 \\ t \neq 2009}}^{2019} \beta_t \mathbf{I}(\text{year} = t) + \alpha_t + \varepsilon_{ijkt} \quad (\text{Model 2})$$

$$inb_{ijkt} = treat_j + treat_j \times \sum_{\substack{t=2005 \\ t \neq 2009}}^{2019} \beta_t \mathbf{I}(\text{year} = t) + \alpha_t + \alpha_k + \mathbf{X}_{it}\delta + \varepsilon_{ijkt} \quad (\text{Model 3})$$

$$inb_{ijkt} = treat_j + treat_j \times \sum_{\substack{t=2005 \\ t \neq 2009}}^{2019} \beta_t \mathbf{I}(\text{year} = t) + \alpha_t + \alpha_k + \mathbf{X}_{it}\delta + (\mathbf{X}_{it} \times post_t)\rho + \varepsilon_{ijkt} \quad (\text{Model 4})$$

Model (2) is a specification that excludes covariates (PTAs) and decomposes the ATT by year-of-birth (yob) cohort. Model (3) incorporates the bull's genetic traits ($\mathbf{X}_{it}$) and firm-level fixed effects, and Model (4) includes interactions of genetic traits and the post-treatment dummy. We include the last specification to account for trait-level trends. We also know that the treatment affects both inbreeding rates and PTAs, but we do not have a structural model of their impact (Lewbel, 2019), so we include them nonlinearly at both levels and in interactions to reduce uncertainty about the functional form of our controls.

5 Results

Figure 8 shows the treatment and control groups' average inbreeding rates and confidence intervals across years. Genomic selection did not immediately increase inbreeding rates; both curves began to diverge substantially after 2011. Inbreeding rates, in turn, began to increase two years later, and both curves diverged significantly from 2012 onwards. Because genomic testing was adopted essentially industry-wide after 2010 (Figure 5), no valid non-adopting counterfactual is available, and our matched design identifies the *differential* effect of genomic testing on inbreeding in popular lines relative to trait-matched, less-popular lines, not its aggregate effect. The control group's own upward trend after 2015 is consistent with genomic testing diffusing into less-popular lines as well; our estimates capture only the *additional* inbreeding concentrated in popular lines, and $\hat{\beta}_t$ should be read as a lower bound on genomic testing's total effect on industry-wide inbreeding.

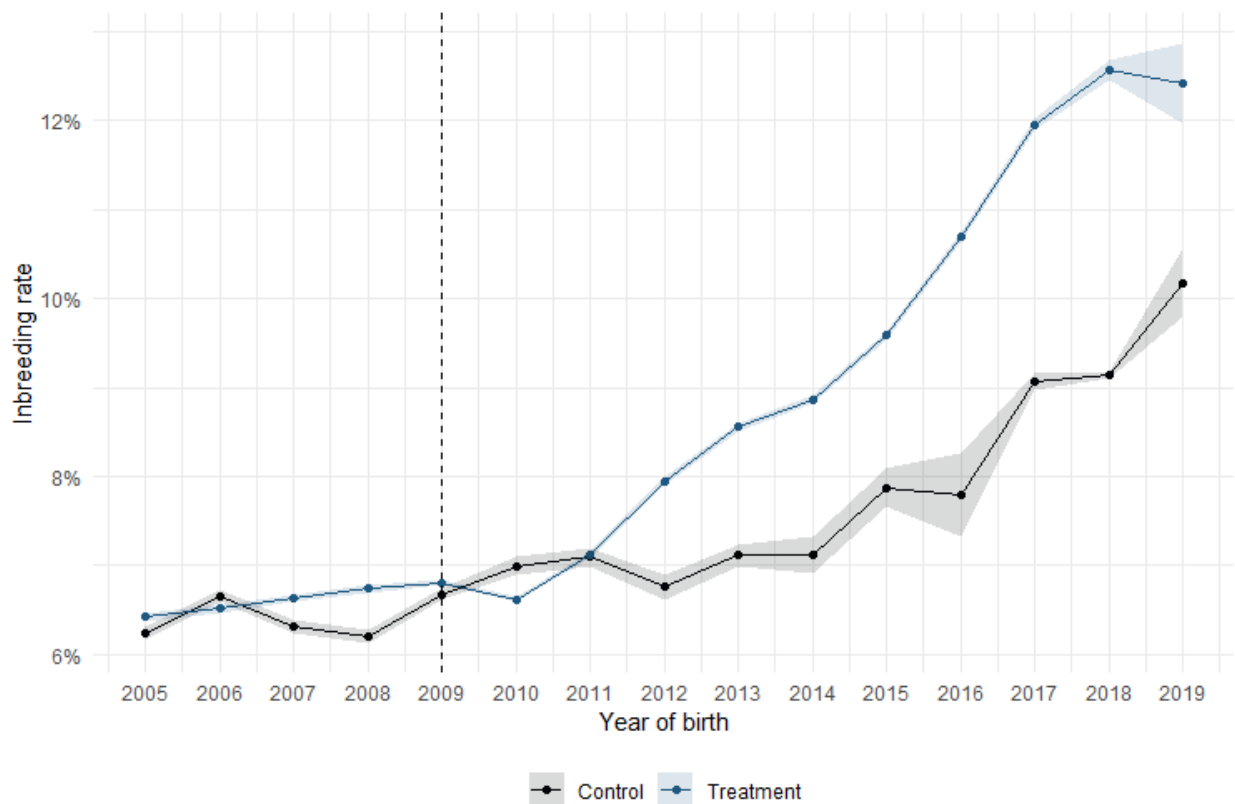


Figure 8: Average inbreeding rates in treatment and control groups

This plot closely matches the evolution of the population-average inbreeding rates shown in Figure 2. This behavior is consistent with our hypothesis that the gains from inbreeding became

higher in these lines after genomic testing was introduced. Our findings align with those reported by Guinan et al. (2023), who also examine the temporal trends in productivity and reproductive traits in US dairy cattle. The authors also document a decrease in the generation interval and an increase in inbreeding rates in Holstein and Jersey cattle, but do not consider the latter an issue because expected inbreeding rates in future offspring adjust genetic trait values.

Table 2 shows the results of three models. Model (1) only includes the interaction between treatment and an indicator for the post period, making it a 2×2 difference-in-differences model. Model (2) splits the treatment effect by year of birth averages, Model (3) includes the complete set of genetic traits and firm-level fixed effects, and Model (4) is the model with traits and a post-treatment dummy. In all specifications, standard errors are clustered at the line level to better account for the inconsistency arising from autocorrelation in inbreeding rates across generations (Bertrand et al., 2004). The number of clusters is reported in Table 2.

Model (1) is the simple differences-in-differences estimation where $post_t$ is a dummy equal to 1 if the bull was born after 2009. The results show that bulls descended from popular bulls increased their inbreeding rates by 1.46 percentage points more than those descended from less popular bulls, on average. This is about a 48% increase relative to the sample mean. Since the impact of genomic testing on inbreeding rates only became noticeable after 2011, the coefficient underestimates the average treatment effect on the treated (ATT) of genomic selection. Model 2 shows that the treatment-year-of-birth interactions are positive and significant from 2012 onwards, with the 2018 effect being the highest. The popular bulls born in 2018 experienced a 3.5 percentage-point increase in inbreeding, meaning they had double the inbreeding rate of their pre-genomics ancestors (2.9%).

Models (3) and (4) in Table 2 show that there is a positive and significant incremental variation for the treated versus control groups in the years 2012-2019 even with the inclusion of controls. In all years after 2010, the coefficients are positive. They both show a positive and increasing inbreeding rate in the treatment group comprised of all descendants from prestige sires relative to the control group of descendants of comparable non-popular sires.⁵

⁵The last period in the sample is 2019 instead of 2020 because we cannot make inferences after the first due to a reduction in the number of animals in the control sample; the number of comparable descendants is too small for us to consider them a valid comparison group. This drastic reduction is a consequence of lower fertility rates in the control group; since those sires fathered fewer sons, their sons will beget even fewer grandsons, and after a few years, this group collapses.

Table 2: Difference-in-Differences estimates from models 1 to 4

	No covariates	inbreeding	
		Traits	Traits and interactions
treat $\times$ post = 1	1.453*** (0.4250)		
treat $\times$ yob = 2005	0.0944 (0.2975)	0.0489 (0.2988)	0.0252 (0.2829)
treat $\times$ yob = 2006	-0.2183 (0.3067)	-0.1182 (0.2933)	-0.2789 (0.2534)
treat $\times$ yob = 2007	0.2328 (0.3038)	0.2826 (0.2676)	0.2848 (0.2534)
treat $\times$ yob = 2008	0.4035 (0.3148)	0.3246 (0.2798)	0.3294 (0.2425)
treat $\times$ yob = 2010	-0.4589 (0.2827)	-0.4721* (0.2825)	-0.5583** (0.2667)
treat $\times$ yob = 2011	-0.0164 (0.3599)	-0.0455 (0.2596)	-0.1671 (0.2511)
treat $\times$ yob = 2012	1.224** (0.4906)	1.252** (0.5633)	1.151* (0.5981)
treat $\times$ yob = 2013	1.433*** (0.4124)	1.083** (0.4631)	0.9980** (0.4725)
treat $\times$ yob = 2014	1.780*** (0.3017)	1.600*** (0.3513)	1.501*** (0.3068)
treat $\times$ yob = 2015	1.637*** (0.3453)	1.363*** (0.4159)	1.255*** (0.4256)
treat $\times$ yob = 2016	2.414*** (0.2424)	1.766*** (0.2697)	1.710*** (0.2564)
treat $\times$ yob = 2017	2.849*** (0.4456)	2.770*** (0.4429)	2.734*** (0.4507)
treat $\times$ yob = 2018	3.417*** (0.4530)	3.576*** (0.4566)	3.515*** (0.4847)
treat $\times$ yob = 2019	2.005** (0.8611)	2.071** (0.8827)	2.022** (0.8785)
p-value for nonzero pre-effect	0.416	0.498	0.107
Number of clusters	185	185	185
Observations	15,491	15,491	15,490
R ²	0.16988	0.47369	0.49938
Within R ²	0.00702	0.01235	0.04150
treat fixed effects	✓	✓	✓
post fixed effects	✓		
yob fixed effects		✓	✓
parent_company fixed effects		✓	✓

Table 2 also includes the results for the F-test for joint significance of all pre-treatment differences in trends for the interaction of treatment and year of birth. The p-values are not significant at any specific level in any of the specifications.⁶

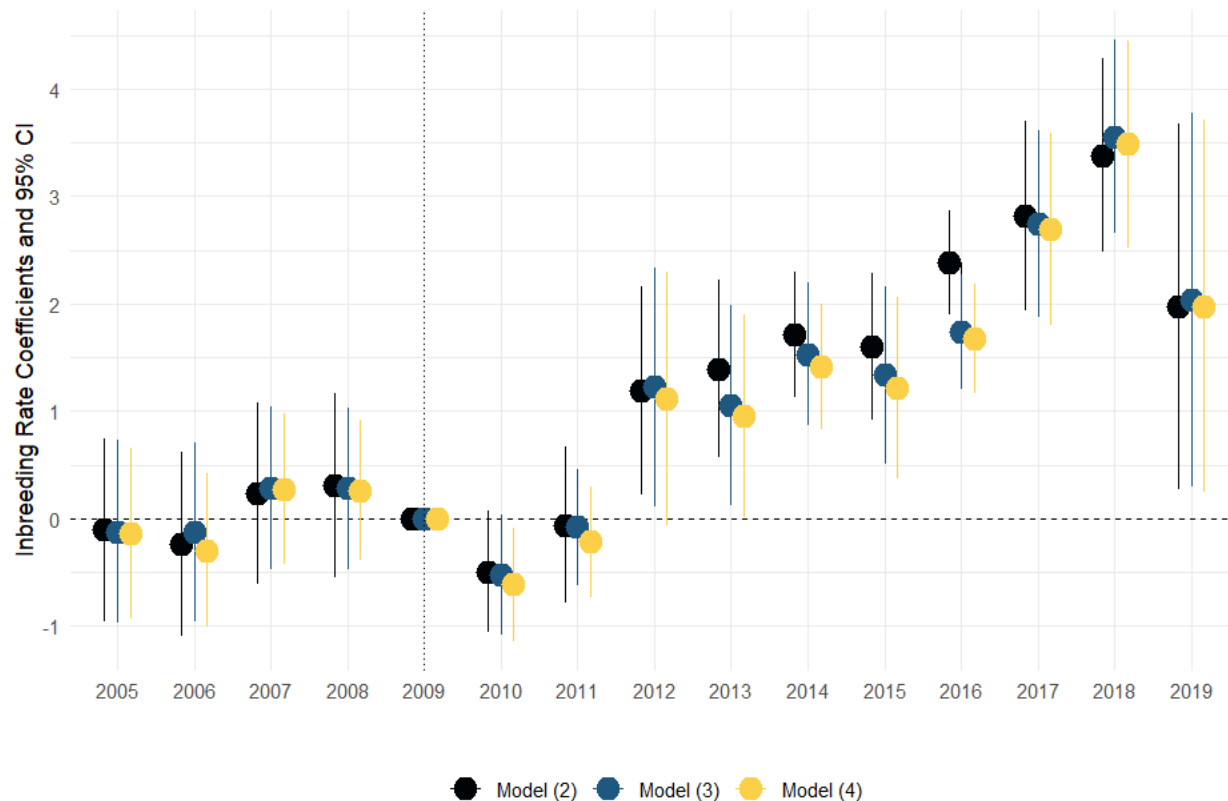


Figure 9: Coefficient plots for different specifications

Figure 9 shows the coefficients from the interaction of year fixed effects and treatment status to assess the impact of inbreeding per year. Every year after 2012, the inbreeding rate of the treatment group is between 1 and 3.5 percentage points higher than the control group. The coefficient values decrease slightly after controlling for their genetic traits, but remain statistically significant. The figure also shows that all pre-treatment coefficients are not significant. Figure 9 uses the *sup-t* uniform confidence intervals for the pre-treatment periods of Montiel-Olea and Plagborg-Møller (2019) as suggested by Freyaldenhoven et al. (2019) instead of the standard pointwise confidence.

⁶This is not a test for parallel trends since the assumption relies on unobservable counterfactuals. Rather, it is usually considered evidence for the assumption of parallel trends. However, this is not necessarily the case since this test has low power against a linear or quadratic pre-trend alternative (Roth, 2022). In our case, we can dismiss this concern: since our matching method ensures the founders meet the condition, the entire effect must be due to their sons and grandsons.

These confidence intervals contain the actual path of the coefficients 95% of the time and can better help understand which types of pre-trends are consistent with the data.

6 Discussion

This article investigates the consequences of genomic selection on the US dairy industry. On the one hand, this new technology significantly improved productivity. On the other hand, we find evidence that it increased the average inbreeding rate of all bulls in the market. As genomic testing increased the rate of genetic improvement, it appeared to exert competitive pressure on animal breeders to select from popular lines with high gains. By not crossing enough with other lines, the inbreeding rate increases, dampening future productivity gains by depleting genetic diversity over the long term. As the average inbreeding rate increases, each new generation of animals faces a higher risk of inbreeding depression and of unpredictable recessive genetic diseases.

It should be noted that the increase in inbreeding we document is not necessarily evidence of harmful behavior by individual breeders. As the referee rightly notes, some degree of inbreeding is both inevitable and strategically rational in purebred selection programs, and its net effects depend heavily on the monitoring and selection practices in place. Our concern is specifically whether genomic testing has pushed the industry beyond the socially optimal level of inbreeding, a question that our empirical results address only indirectly. A full welfare analysis would require knowledge of the social optimum d^* , which is outside the scope of this paper. The USDA publishes a profitability index called Net Merit that estimates the profitability contribution of a given dairy bull based on its genetic traits. Even after correcting for inbreeding depression, we estimate that the index increased by 275 USD per animal relative to what it might have been if genetic traits had not increased after 2010.⁷

The highest per-animal cost of inbreeding in our findings is about 140 USD, compared with lines that are not popular. These costs are difficult to compare since one is relative to a control group and the other is not.⁸ Regardless, it appears that the net benefits of genomic testing are positive in the short term. This short-run comparison omits the direct cost of genomic testing itself

⁷Estimated as the 2011–2020 average deviation of the Net Merit index from an ARIMA projection fit to its 1985–2009 values, i.e. the per-animal gain relative to the pre-genomic-testing trend.

⁸See Appendix A for more details on our calculations.

(genotyping expenditure and any associated replacement or culling losses which our data do not measure). A full benefit–cost analysis of genomic testing is outside the scope of this paper. We note, however, that the cumulative discounted genotyping cost over our window is of a substantially smaller order of magnitude than the estimated productivity benefit, so its omission narrows but does not reverse the net-positive result.

While there are clearly short-term gains to genomic testing, the actual welfare effects are harder to discern. As we explore in our conceptual framework, there may still be welfare losses relative to the counterfactual of no genomic testing. However, our paper does not aim to examine such a counterfactual; instead, it focuses on the relative increase in inbreeding among popular lines. This is because reducing genetic diversity through inbreeding increases short-run genetic gain at the cost of steadier long-run productivity growth. At its worst, excessive inbreeding leads to a population collapse, leaving no genetic diversity to sustain continued production.

Because dairy cattle breeders are not incentivized to cross-breed in the current market, this suggests a role for the government or an industry group to help address the coordination problem. Currently, the Holstein Association of the United States adjusts genetic trait values based on inbreeding levels to penalize breeders who select bulls with high inbreeding, reflecting the higher reproductive costs of such animals. Yet, Cole (2024) points out that many of the most inbred bulls have such high levels of traits that the downgrading will not temper the demand for them enough. Another option would be to give dairy farmers more information about the dangers of inbreeding to reduce demand for the most inbred lines (Cole, 2024).

More research is needed in this area of agricultural economics, as the coordination problems in dairy cattle breeding have not yet been examined in the field. Future work could make a significant contribution by expanding our conceptual framework to understand what the socially optimal level of inbreeding is and under what circumstances a technology like genomic testing actually deepens the welfare losses from inbreeding. A full cost-benefit analysis of genomic testing is, unfortunately, outside the scope of this paper. Yet, studying the increase in inbreeding is crucial for ensuring a safe, globally resilient dairy sector and for ensuring that US dairy cattle breeders continue to sustainably raise dairy cattle productivity.

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A Genomic Selection Cost Appraisal

Using published estimates of the costs of inbreeding, we can assign a monetary value to increases in inbreeding rates. The Council of Dairy Cattle Breeding (CDCB) estimates that each additional percentage of inbreeding is associated with a 40.11 USD reduction in income from dairy product sales. The total cost depends on the average inbreeding rate we consider.

We can use the coefficients from Model (4), our preferred specification, as the conditional difference in inbreeding rates between the control and treatment groups. In other words, we can assign monetary values to the Average Treatment on the Treated values (ATT) in our sample. We cannot extrapolate this result to the entire set of bulls, since we did not choose our sample at random. The per-animal average increased cost in year t is equal to:

$$\overline{Cost}_t = 40.11 \times \hat{\beta}_t \quad (2)$$

Because our ATT measures the *differential* cost of inbreeding in popular lines relative to matched controls (Section 4), we set N_t equal to the number of registered descendants of popular founder lines born in year t , the treated population to which the ATT applies, rather than the total US Holstein population. This yields a partial accounting of the inbreeding costs induced by genomic testing: it excludes the concurrent rise in inbreeding among non-popular lines (Figure 8), which our design differences out. The resulting figure is best understood as the cost of the popularity-specific inbreeding premium, and a conservative component of the industry's total inbreeding cost. Because the treated (popular-line) share of the national herd is not directly observed, we report the cost across a range of assumed shares. Finally, we calculate the net present value using a discount rate (r) equal to 4.5%:

$$\begin{aligned} NPV &= \sum_{t=2012}^{2019} \frac{1}{(1+r)^{t-2012}} N_t \times \overline{Cost}_t \\ &= \sum_{t=2012}^{2019} \frac{1}{(1+r)^{t-2012}} N_t \times (40.11 \times \hat{\beta}_t) \end{aligned}$$

Table 3 reports the resulting NPV (2012 base year). Applied only to the popular-line descendants we observe, the cost is a firm floor of about USD 0.4 million and involves no extrapolation. Scaling the same per-animal differential to the fraction of the national Holstein herd descended from popular lines yields USD 1.3–3.0 billion at a 4.5% discount rate for shares between 30% and 70%. Wuepper et al. (2023) document that farmers in North America and Western Europe apply higher discount rates than the market rate, around 23% per annum; at that rate the same figures fall to roughly USD 0.6–1.4 billion. These magnitudes reflect the costs on one side of a heterogeneous market and should be read as illustrative, assuming a given share.

Table 3: Net present value of the popularity-specific inbreeding cost, 2012–2019 (2012 base year)

Scaling of treated population	Discount rate	
	$r = 4.5\%$	$r = 23\%$
Treated animals only (no extrapolation)	\$0.4 M	\$0.2 M
<i>Share of national herd descended from popular lines:</i>		
30%	\$1.27 B	\$0.60 B
40%	\$1.69 B	\$0.81 B
50%	\$2.11 B	\$1.01 B
60%	\$2.53 B	\$1.21 B
70%	\$2.95 B	\$1.41 B

Notes: Per-animal cost is $40.11 \times \hat{\beta}_t$ (Model 4). “Treated animals only” applies the differential to the popular-line descendants in our sample and involves no extrapolation. The population-scaled rows apply the same differential to the stated fraction of the national Holstein herd and should be read as illustrative under an assumed share.

B Benefits from Genomic Selection

While genomic testing increased inbreeding, it also increased milk production and, in turn, profits for the dairy industry.

The benefits of introducing genomic selection in our sample can be calculated using an index called **Net Merit**, which measures the net profit over the lifetime of the average daughter of the animal, expressed in U.S. dollars. Net merit is calculated by multiplying the values of genetic traits by a set of dollar values intended to capture the “market value” of all traits.⁹

Published values of Net Merit are already adjusted for inbreeding; the increase in quality will exceed the losses from inbreeding depression. A more accurate estimate of the benefits requires

⁹Table 6 shows what are the magnitudes of such weights (Van Raden et al., 2021). Note that the calculation involves the usage of subindexes for health and reproduction traits, as well as an estimation of residual feed intake per animal.

more information than we have; we need information on the expected future inbreeding for all animals in our sample. We do not have the information required to perform such a calculation.¹⁰

In principle, one could compare the Net Merit of popular-line descendants with that of matched controls to gauge whether the extra genetic gain in popular lines offsets their higher inbreeding cost. In practice, our design does not identify this net welfare effect. The per-year Net Merit differential between popular and control lines is small relative to its sampling variability (driven by the limited size of the matched control group) and switches sign across birth cohorts. A line-clustered bootstrap of the benefit–cost ratio yields a 95% interval that spans values well below and well above one (equivalently, a net present value whose interval comfortably contains zero), so we do not report a net welfare figure. We therefore restrict our welfare accounting to the inbreeding *cost* of the popularity premium, which is well identified (Appendix A), and leave estimation of the net welfare effect to future work with larger control samples.

C Additional Estimator Assumptions

Our causal estimand of interest is the Average Treatment on the Treated (ATT) in the post-intervention period:

$$ATT_{post} = E \left[inb_{post}^{treat} - inb_{post}^{control} \middle| T_j = 1 \right]$$

This quantity measures the average causal effect on the treated lines in the post-treatment period (Roth et al., 2023). To correctly identify causal effects, a series of assumptions must hold (Sun and Abraham, 2021):

Assumption 3 (Parallel trends):

$$E \left[inb_{post}^{control} - inb_{pre}^{control} \middle| T_j = 1 \right] = E \left[inb_{post}^{control} - inb_{pre}^{control} \middle| T_j = 0 \right]$$

We assume that the average inbreeding rates for both groups would have evolved in parallel without introducing genomic selection. This assumption is not testable because we cannot observe

¹⁰Let g_k be the reported PTA of trait k from bull i and $\tilde{g}_{ik}$ be the unadjusted PTA, then their relationship will be: $\tilde{g}_{ik} = g_{ik} - [(EFL_i - F_i)\hat{\beta}_k]$, where EFL_i is the expected future inbreeding and $\hat{\beta}_k$ is the regression coefficient from a linear model that relates observed values with phenotypic values.

untreated inbreeding rates after 2009, but it constrains our model specification to a Two-Way Fixed Effects specification we call Model (1):

$$inb_{ijkt} = treat_j + post_t + (treat_j \times post_t)\beta + \varepsilon_{ijkt} \quad (\text{Model 1})$$

Where inb_{ijkt} is the inbreeding rate of bull i from line j , supplied by firm k and born in year t ; $treat_j$ indicates whether line j is defined as the difference-in-differences in observed outcomes between the pre- or post-2009 period, and ε_{ijkt} is the error term. Firm fixed effects enter Models 3 and 4, not Model 1.

Assumption 4 (No anticipatory effects): $inb_{pre}^{control} = inb_{pre}^{treat}$

This assumption ensures that both groups were comparable before the treatment; inbreeding rates of popular lines did not increase before the introduction of genomic selection; we use matching methods to ensure that this condition holds.

Under Assumptions 3 and 4, we have that:

$$ATT_{post} = E \left[inb_{post}^{treat} - inb_{post}^{control} \middle| T_j = 1 \right] - E \left[inb_{pre}^{treat} - inb_{pre}^{control} \middle| T_j = 0 \right] \quad (3)$$

The ATT is defined as the difference-in-differences in observed outcomes between the pre- and post-intervention periods. All elements in this equation are observed and, therefore, estimable. Suppose popular bulls had superior traits to our matched control group. In that case, this condition is violated, and we would not be able to parse out the causal effect of inbreeding from the effect of the bull's genetic superiority. Assumptions 3 and 4 ensure that the OLS estimates $\hat{\beta}$ from Model (2) are consistent and asymptotically valid (Roth et al., 2023).

Assumption 5: Let $W_j = (inb_{j,pre}, inb_{j,post}, T_j)'$ be a vector of outcomes and treatment status for unit i . Then, we observe a sample of N independent and identically distributed random draws $W_i \sim F_W(w)$ for a distribution F satisfying parallel trends.

Under Assumptions 3, 4, and 5: $\sqrt{N}(\hat{\beta} - ATT_{post}) \xrightarrow{D} N(0, \sigma^2)$, then we can consistently estimate the variance σ^2 clustering at the line level (Bertrand et al., 2004) asymptotically ($N \rightarrow \infty$ and T fixed).

We can also decompose the variation in treatment time, since each year a certain number of

bulls will be born from lines in both groups, each with its own inbreeding rate that can be averaged across the treatment and control groups. Models 2-4 show the complete specification, including leads and lags, for the treated group. These models decompose the estimated ATT relative to the treatment average by birth year. Sun and Abraham (2021) call this specification “Dynamic specification.” However, there is no staggered treatment: genomic selection affects all lines in the same year, whereas inbreeding rates evolve independently thereafter.

This assumption states that an individual’s treatment status on any given line is unaffected by the treatment status of others. In our setting, the treatment (genomic selection) affects all individuals, so there is no selection into treatment.

D Robustness Checks

D.1 Alternative Matching Estimators

Our baseline constructs the control group by nearest-neighbor matching on a logit propensity score. To assess sensitivity to this choice, we re-estimated the matched control group under seven alternative estimators: Mahalanobis-distance matching, propensity-score matching with a caliper of 0.2 standard deviations, one-to-one matching without replacement, optimal pair matching, full matching, inverse-probability weighting (IPW), and entropy balancing.

Table 4 reports, for each method, the number of covariates (of 74) achieving an absolute standardized mean difference below 0.1, the largest remaining difference, and the effective sample size (ESS) of the matched control group.

Three of the alternatives are not feasible in our setting, and the reason is instructive. With only 40 treated founders and 74 matching covariates, methods that rely on a high-dimensional distance metric or on exact moment balancing break down. Mahalanobis matching balances only 32 of 74 covariates, as Euclidean distance loses discriminating power in high dimensions. Full matching and entropy balancing achieve worse balance than the baseline while driving the effective control sample toward degeneracy: entropy balancing places essentially all weight on a single control unit (control ESS ≈ 1) yet still leaves 49 covariates imbalanced, and full matching collapses the control ESS to below nine. We therefore do not report these three methods as robustness checks; their failure reflects the dimensionality of the balancing problem relative to the number of treated units,

not a deficiency of the design.

Among the feasible estimators, the results reinforce the baseline. Caliper matching achieves the best balance of any method (67 of 74 covariates below the 0.1 threshold, improving on the baseline’s 56), at the cost of a smaller matched treated group, since the caliper discards superstar founders with no sufficiently close control (treated ESS falls from 40 to 29). Inverse-probability weighting balances 59 of 74 covariates while retaining a comparable control sample ($ESS \approx 62$). In both cases, the balance is at least as good as the baseline without degeneracy.

Most importantly, the difference-in-differences estimate is stable across these specifications. Re-estimating Model (1) on the caliper-matched sample yields a treatment-by-post coefficient of 1.19 percentage points (0.423), and the IPW-weighted specification yields 0.98 (0.458), both of similar magnitude to the baseline estimate of 1.45 percentage points (0.607) and statistically significant at conventional levels.

Figure 10 overlays the event-study coefficients under the baseline, caliper, and IPW specifications; the post-2011 divergence in inbreeding between popular and control lines is present and of similar magnitude in all three. We conclude that our findings are not an artifact of the propensity-score matching estimator.

Table 4: Covariate balance and effective sample size across matching estimators

Estimator	Balanced (< 0.1), of 74	Largest SMD	Control ESS
<i>Reported</i>			
Nearest neighbor (baseline)	56	0.28	76.4
Caliper (0.2 SD)	67	0.18	44.0
Inverse-probability weighting	59	0.20	62.0
<i>Infeasible at 74 covariates / 40 treated (not reported)</i>			
One-to-one, no replacement	50	0.36	40.0
Optimal pair matching	38	0.65	360.9
Mahalanobis distance	32	0.79	111.1
Full matching	38	1.15	8.6
Entropy balancing	25	2.36	1.0

Notes: Balance is the count of the 74 matching covariates with absolute standardized mean difference below 0.1; “Largest SMD” is the maximum absolute standardized mean difference. Effective sample size (ESS) is for the matched or weighted control group; the treated group has 40 founders (29 under the caliper). Standardization uses the treated-group standard deviation (ATT). Mahalanobis, full, and entropy balancing are shown for completeness but not used, as high-dimensional distance and exact moment balancing are infeasible with 40 treated units.

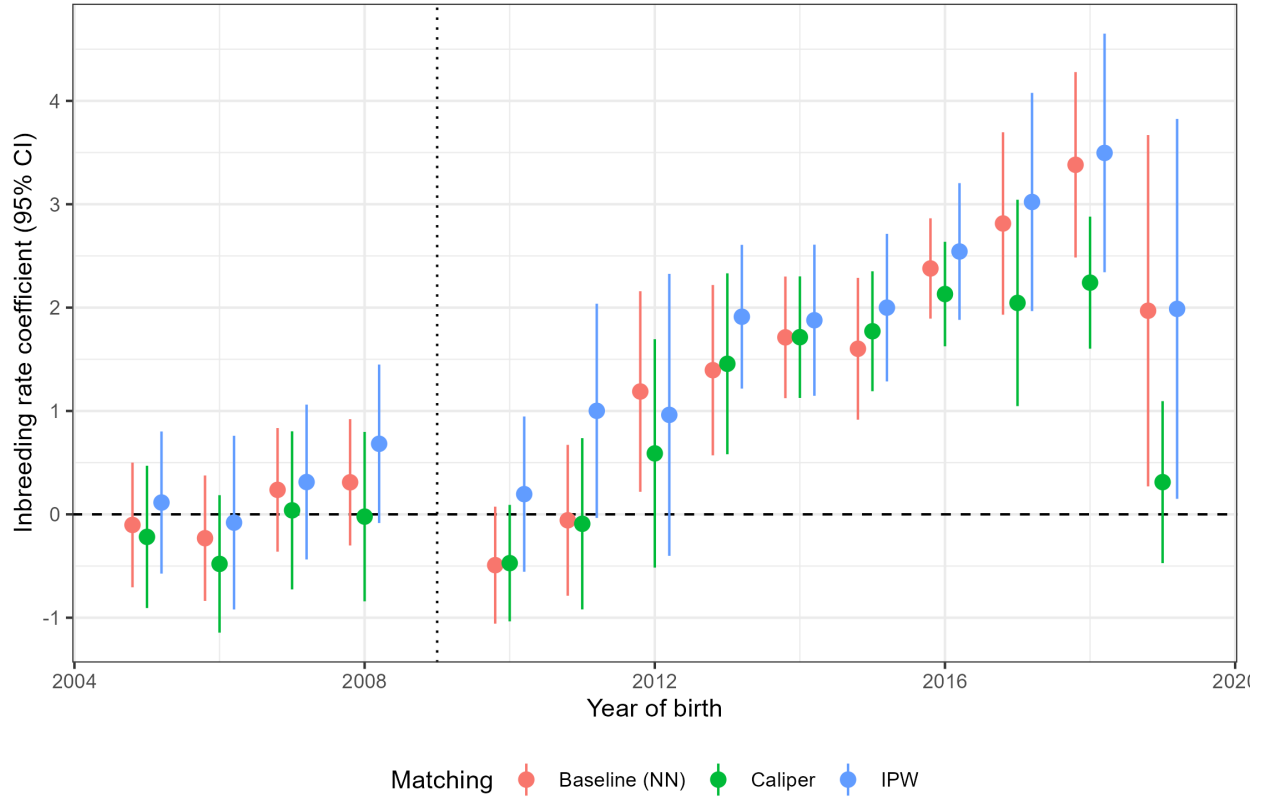


Figure 10: Coefficient plot with different matching algorithms

D.2 Total number of descendants

In our analysis, we defined a “popular bull” as a bull whose total number of descendants was among the 95th percentile of their distribution by 2021. The cutoff point also influences this calculation, since genomic testing expands the supply of bulls, and the size of any line is endogenous to genomic selection. It can be argued that selecting bulls based on the ex-post number of sons is tautological with respect to our argument.

Figure 11 plots the coefficients for the different differences-in-differences specifications, but this time, the number of descendants is calculated only up to 2010, so we assume that genomic selection is just being introduced, and we are trying to guess which lines will be more successful. Coefficients for 2014 and 2015 are now substantially reduced and become insignificant, but the overall result remains unchanged; we still have that the treatment group has notoriously higher inbreeding rates than the control group, as seen in the 2×2 model, where the coefficient is still positive and significant for the entire period, as seen in 7.

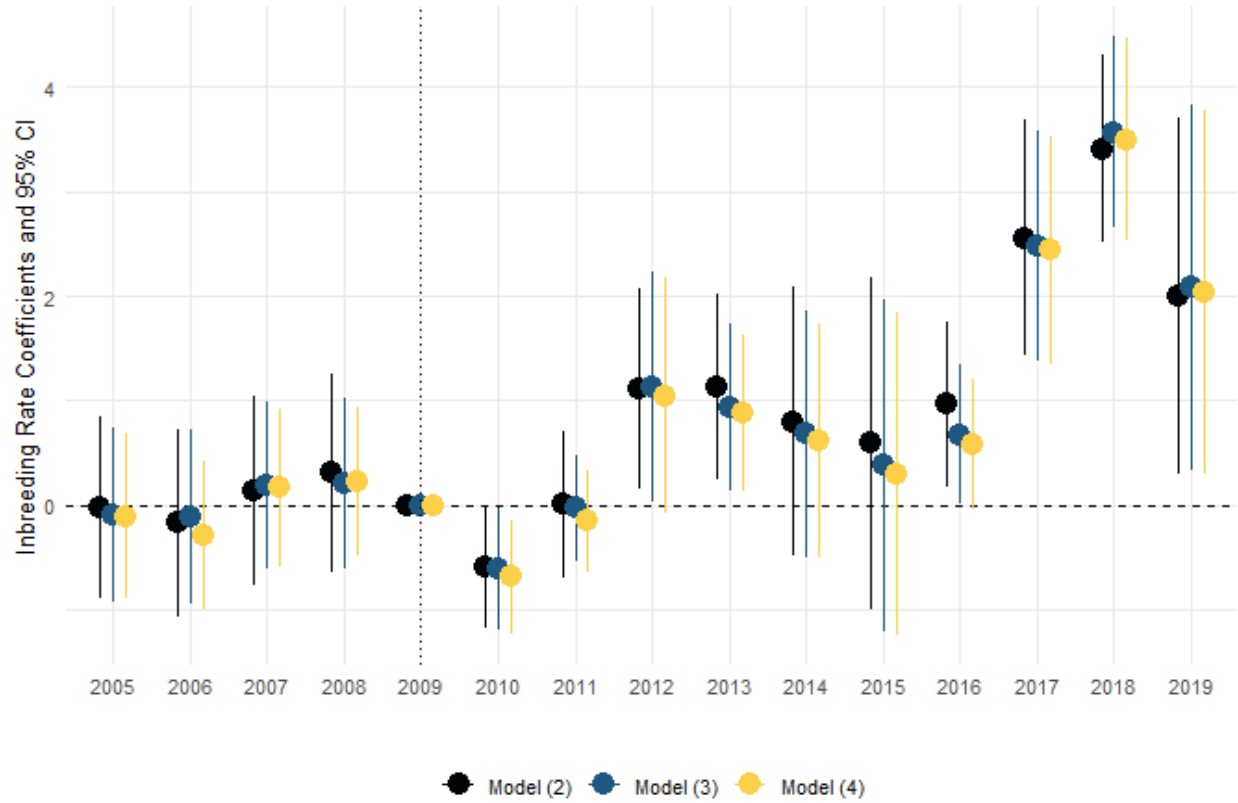


Figure 11: Coefficient plots for an alternative definition of popular bull

D.3 Quarter of birth effects

Since we have the exact date of birth for every animal in the sample, we can also group them by quarter of birth; Figure 12 shows the results from Models 2-4, replacing the year with quarter-of-birth interactions with the treatment. The signs and magnitudes of the coefficients do not differ substantially from those in the original formulation. The increase in coefficient values has been monotonic and statistically significant since 2012, but the smaller quarterly sample size makes confidence intervals wider.

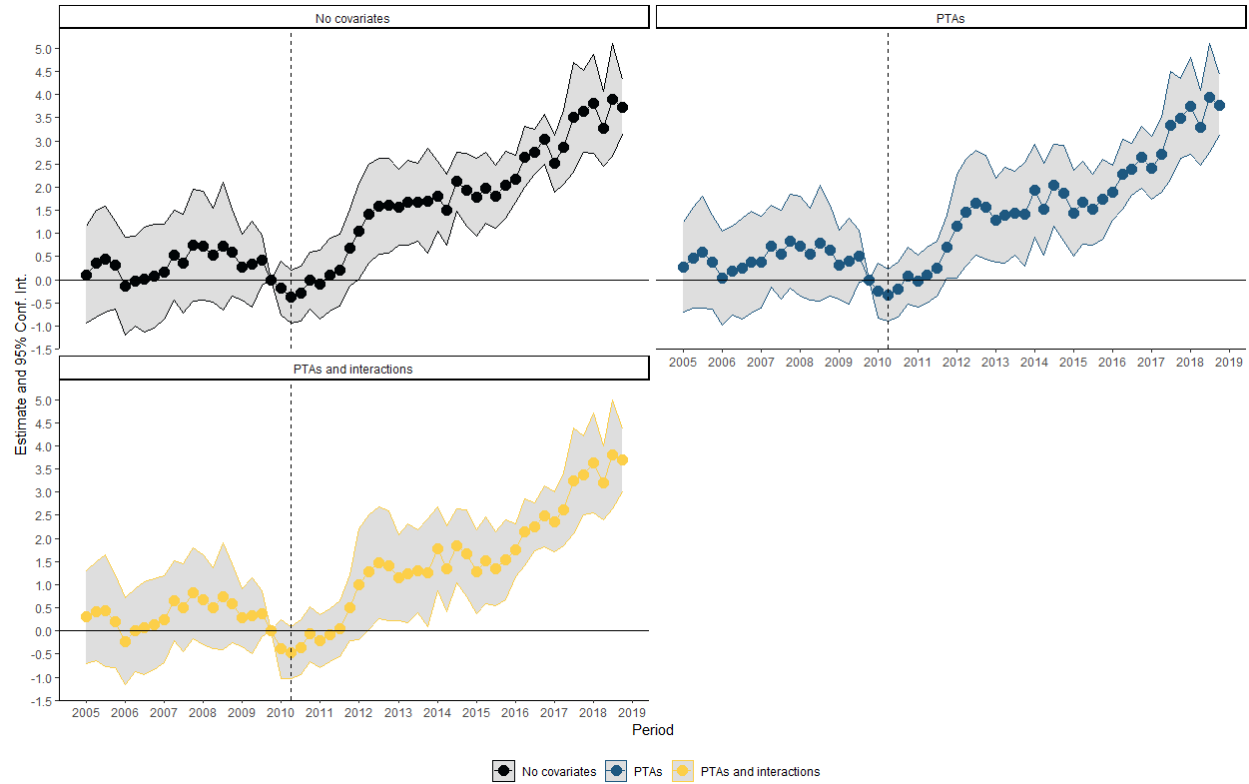


Figure 12: Coefficient plots for quarter of birth effects

This article’s results depend heavily on a somewhat arbitrary definition of a “popular” bull in terms of the size of his line. What would happen if we changed that from the 95th to another quantile in the line-size distribution? Figure 13 shows the coefficient plots for different definitions of “popular”.

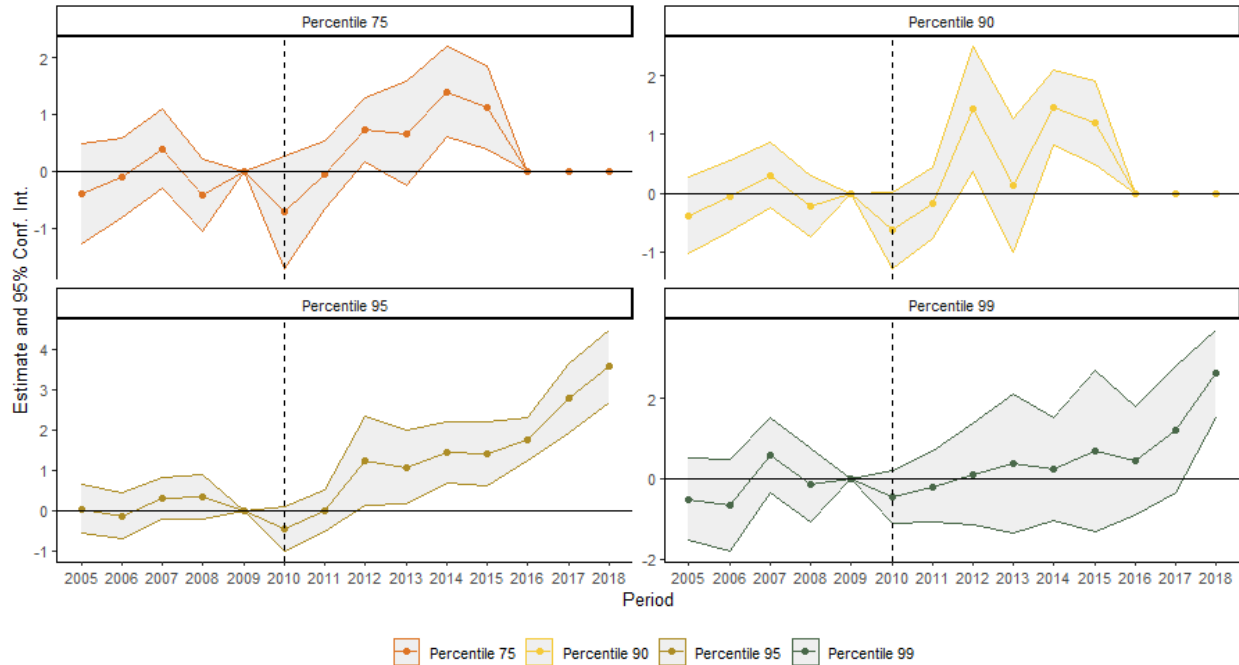


Figure 13: Robustness test: changes in the definition of "popular"

If we change the cutoff point from the 95th to the 99th percentile, the sign and magnitude of our coefficients do not change much. Still, the confidence intervals widen significantly because our treatment group comprises such specific animals that the matching algorithm cannot find suitable matches. As a consequence, the significance of the difference-in-differences coefficients decreases significantly. Going in the other direction, we can find marginally better matches for the popular group. However, the control group ended abruptly in 2015 because most of the original group's control lines are now included in the treatment, making the control group short-lived. We conclude that our results depend critically on a set of lines that allow for a good comparison when included in the control group; otherwise, the analysis yields insignificant results.

D.4 Date of birth of popular bulls

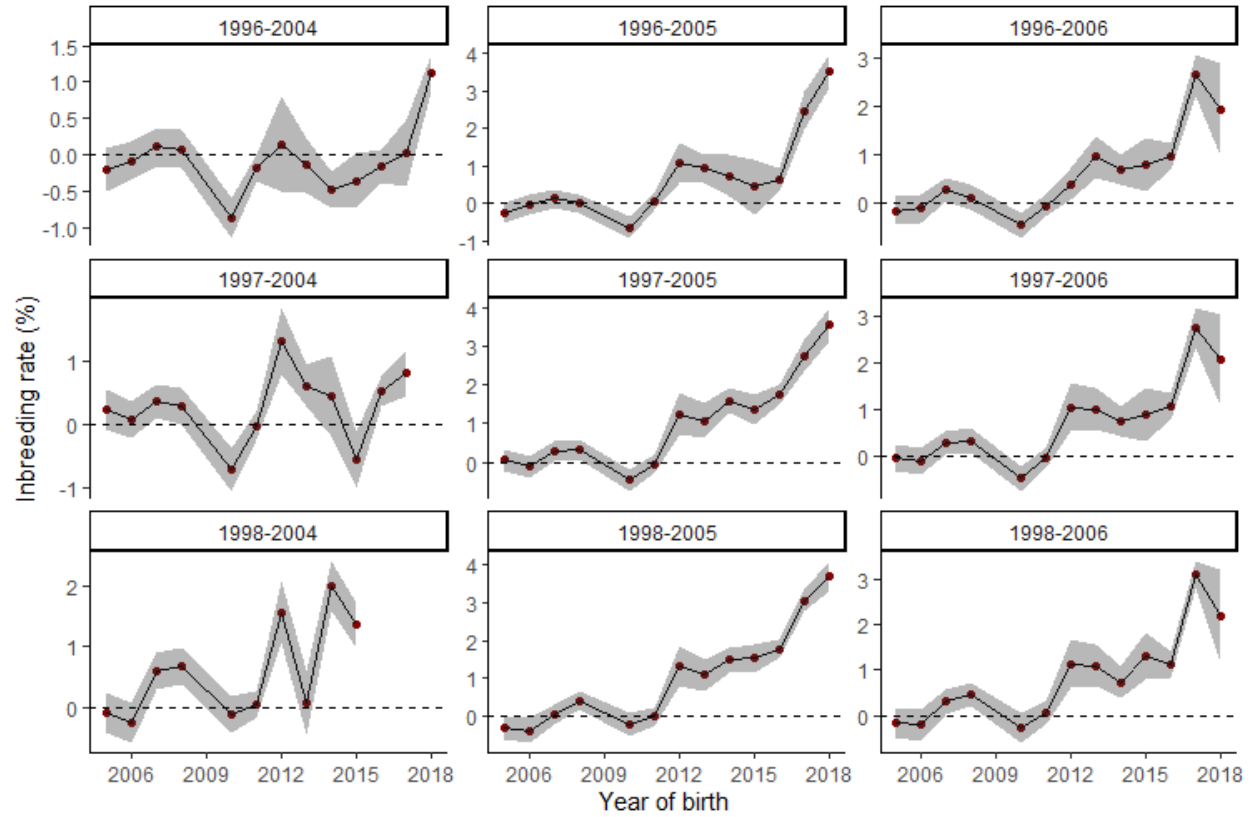


Figure 14: Coefficients from model (2) per different definitions of founders' date of birth

Finally, Figure 14 shows coefficient plots by different definitions of the founders' date of birth. We defined founders as all bulls born between 1997 and 2005, but such a definition is also arbitrary. If we adopt different cutoffs for the period when the founders were born, from 1996/1998 to 2004/2006, we have $3 \times 3 = 9$ distinct combinations for the start and end dates. The coefficient plot shows that the bulk of the effect can be attributed to sires born from 1996 to 2006; if we extend further the selection period of sires, we will have more endogeneity issues since, by this date, it was known that the bovine genotype was being sequenced, and thus, that there would be a significant effect on the market.

Table 5: Difference-in-Differences estimates from models 1 to 4, using two-way clustering

	No covariates		inbreeding	
	(1)	(2)	Traits (3)	Traits and interactions (4)
treat $\times$ post = 1	1.453*** (0.3835)			
treat $\times$ yob = 2005		0.0944 (0.2673)	0.0489 (0.3287)	0.0252 (0.2863)
treat $\times$ yob = 2006		-0.2183 (0.2605)	-0.1182 (0.2418)	-0.2789 (0.2168)
treat $\times$ yob = 2007		0.2328 (0.2653)	0.2826 (0.2268)	0.2848 (0.2325)
treat $\times$ yob = 2008		0.4035 (0.2766)	0.3246 (0.2691)	0.3294 (0.2333)
treat $\times$ yob = 2010		-0.4589 (0.2711)	-0.4721* (0.2553)	-0.5583** (0.2404)
treat $\times$ yob = 2011		-0.0164 (0.3913)	-0.0455 (0.3461)	-0.1671 (0.3430)
treat $\times$ yob = 2012		1.224** (0.4998)	1.252** (0.5625)	1.151* (0.5882)
treat $\times$ yob = 2013		1.433*** (0.4299)	1.083** (0.4956)	0.9980* (0.5077)
treat $\times$ yob = 2014		1.780*** (0.3256)	1.600*** (0.3654)	1.501*** (0.3079)
treat $\times$ yob = 2015		1.637*** (0.4425)	1.363*** (0.4464)	1.255** (0.4482)
treat $\times$ yob = 2016		2.414*** (0.4178)	1.766** (0.6123)	1.710** (0.6431)
treat $\times$ yob = 2017		2.849*** (0.4665)	2.770*** (0.4543)	2.734*** (0.4625)
treat $\times$ yob = 2018		3.417*** (0.4166)	3.576*** (0.4422)	3.515*** (0.4841)
treat $\times$ yob = 2019		2.005** (0.8292)	2.071** (0.8745)	2.022** (0.8532)
p-value for nonzero pre-effect		0.416	0.498	0.107
Number of clusters	257	257	257	257
Observations	15,491	15,491	15,490	15,490
R ²	0.16988	0.47369	0.49938	0.50467
Within R ²	0.00702	0.01235	0.04150	0.05163
treat fixed effects	✓	✓	✓	✓
post fixed effects	✓			
yob fixed effects		✓	✓	✓
parent_company fixed effects			✓	✓

E Additional Plots

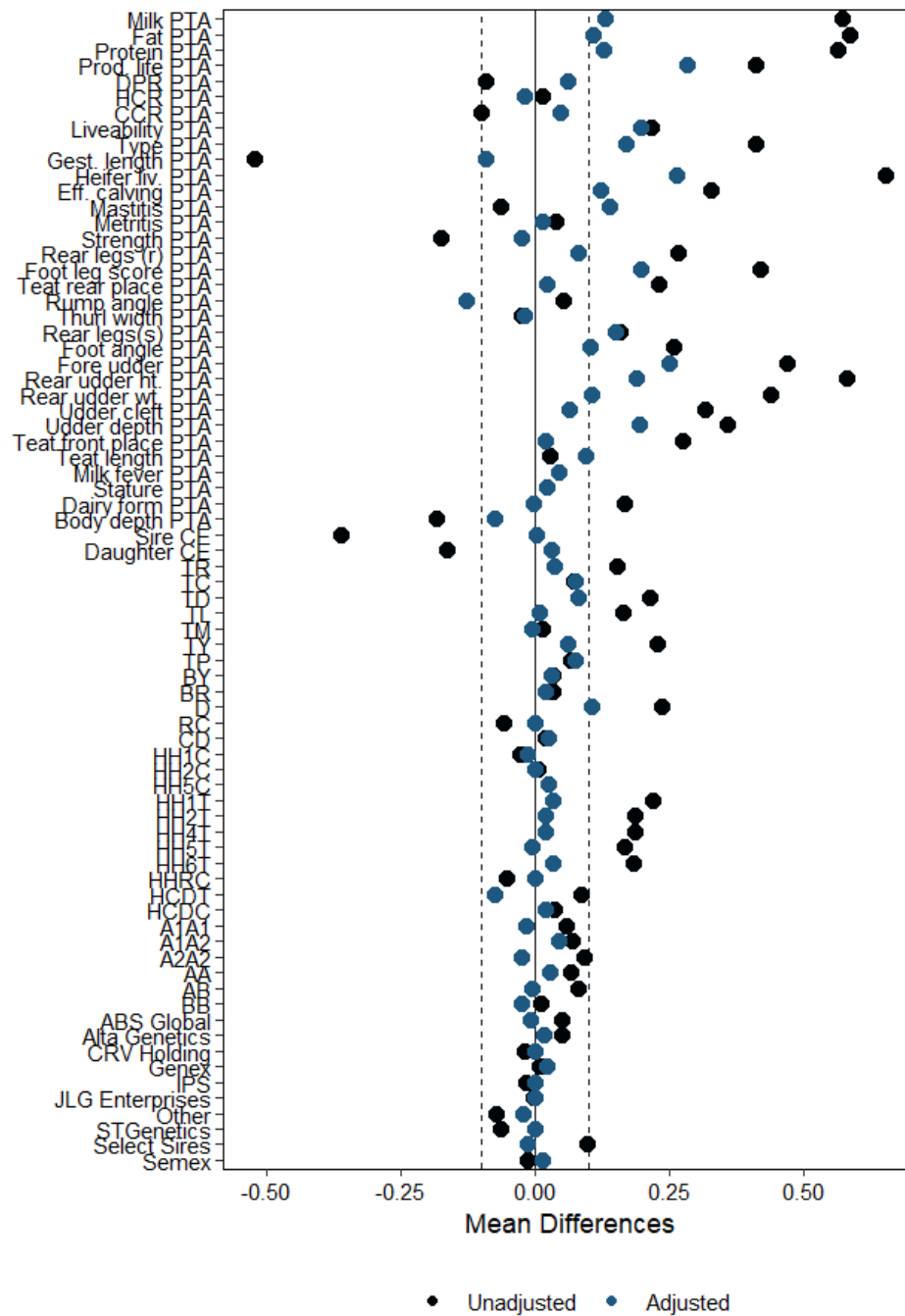


Figure 15: Covariate balance plot

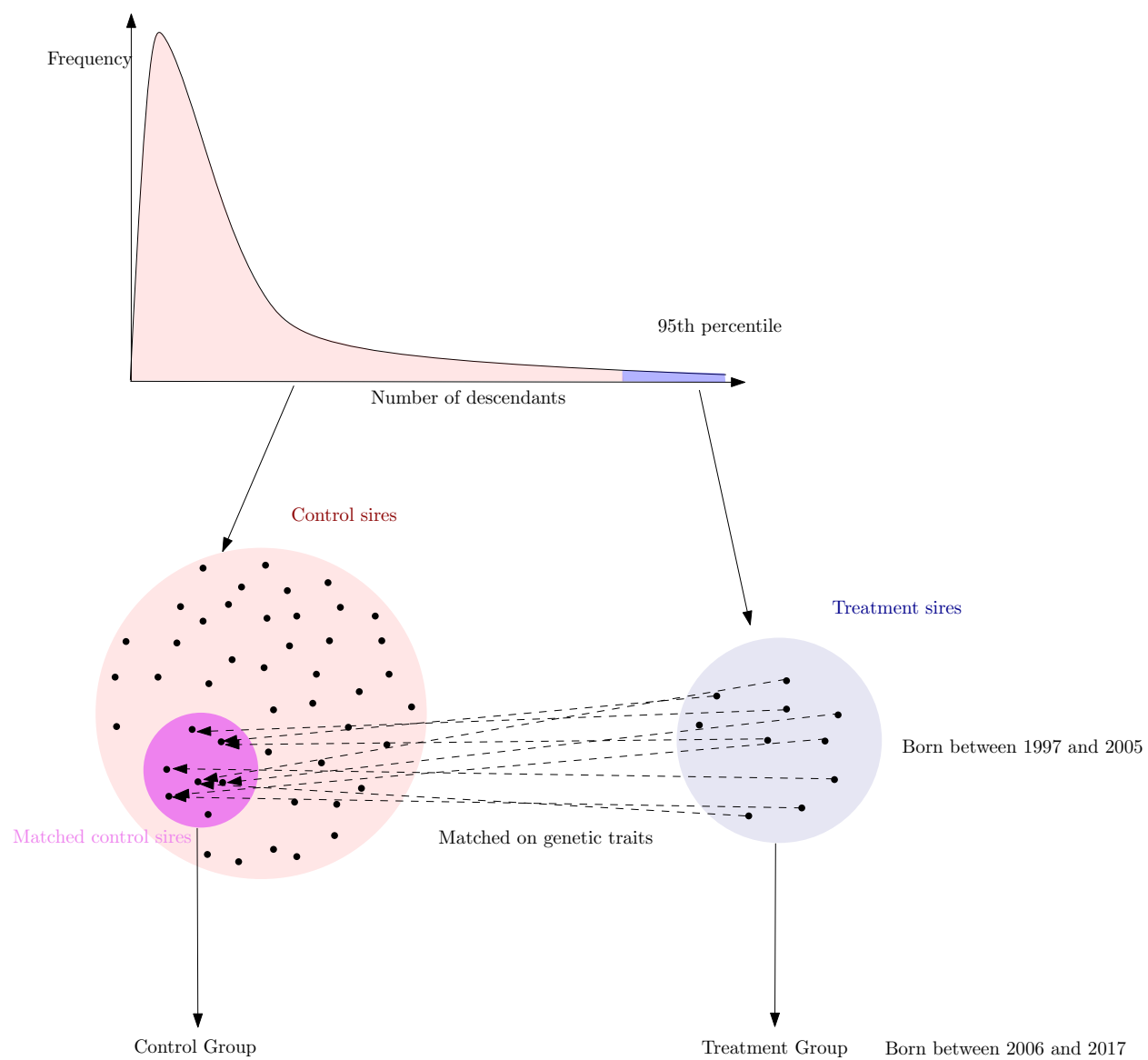


Figure 16: Sample selection mechanism

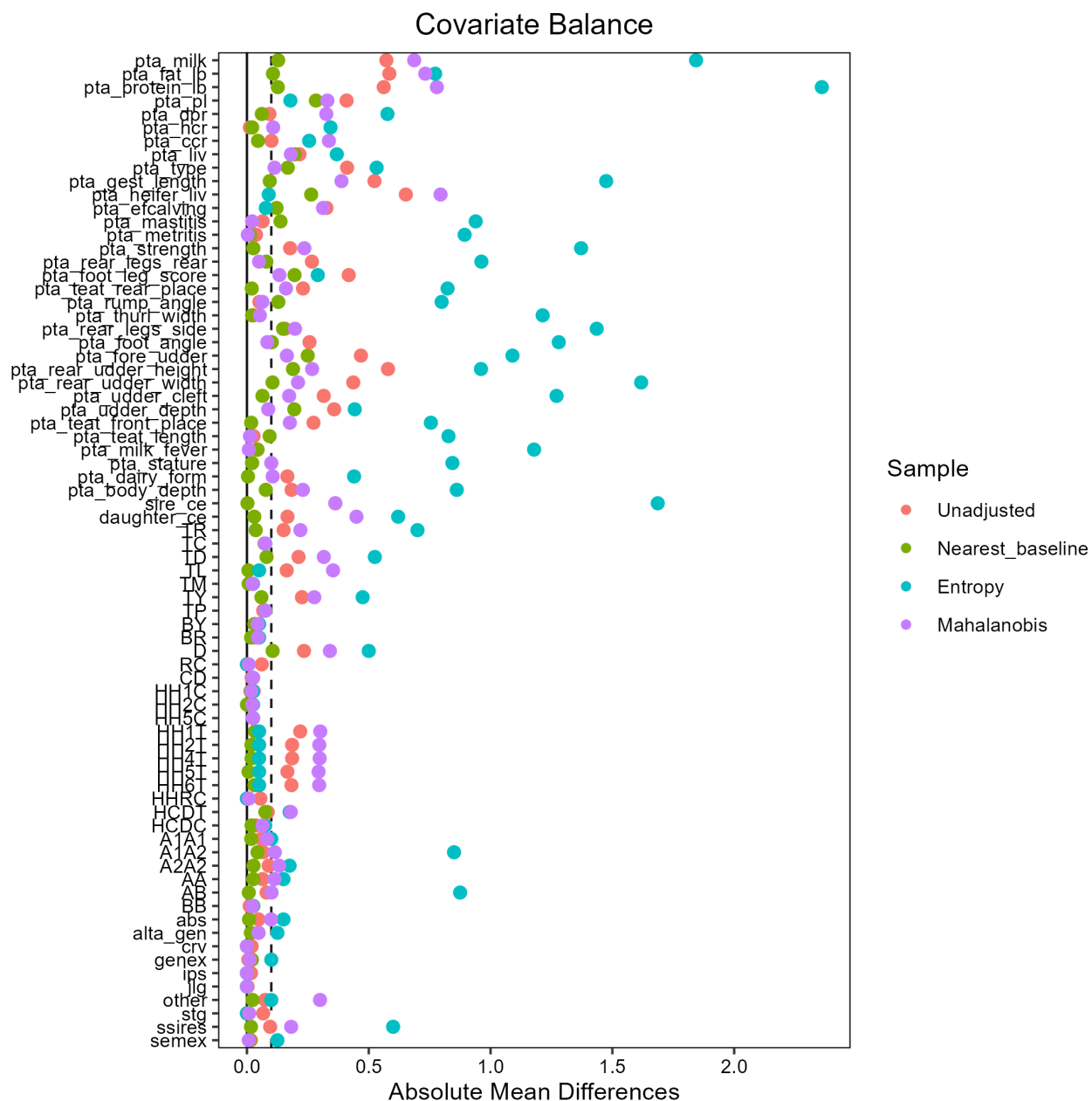


Figure 17: Covariate balance plot after different matching methods

Trait	Units	Value (\$/PTA unit)
Milk	Pounds	0.002
Fat	Pounds	4.18
Protein	Pounds	4.67
Productive Life	Months	34
Somatic Cell Score	Log	-74
Body Weight Composite	Composite	-45
Udder Composite	Composite	19
Foot/Leg Composite	Composite	3
Daughter Pregnancy Rate	Percent	11
Calving Trait Subindex	USD	1
Heifer Conception Rate	Percent	1.1
Cow Conception Rate	Percent	2.2
Livability	Percent	9.8
Health Trait Subindex	USD	1.0
Residual Feed Intake	Pounds	46.2
Effective Calving	Days	2.1
heifer Livability	percent	5.0

Table 6: Economic values for **Net Merit**

Table 7: Difference-in-Differences estimates from alternative scenario

	No covariates		inbreeding	
	(1)	(2)	Traits (3)	Traits and interactions (4)
treat $\times$ post = 1	1.217*** (0.4671)			
treat $\times$ yob = 2005		-0.0189 (0.2991)	-0.0863 (0.2966)	-0.1019 (0.2842)
treat $\times$ yob = 2006		-0.1702 (0.3088)	-0.1060 (0.2951)	-0.2813 (0.2520)
treat $\times$ yob = 2007		0.1376 (0.3131)	0.1949 (0.2826)	0.1698 (0.2699)
treat $\times$ yob = 2008		0.3104 (0.3286)	0.2154 (0.2896)	0.2295 (0.2508)
treat $\times$ yob = 2010		-0.5930** (0.2916)	-0.6049** (0.2935)	-0.6837** (0.2705)
treat $\times$ yob = 2011		0.0069 (0.3567)	-0.0263 (0.2577)	-0.1519 (0.2509)
treat $\times$ yob = 2012		1.115** (0.4877)	1.134** (0.5545)	1.052* (0.5685)
treat $\times$ yob = 2013		1.131** (0.4479)	0.9319** (0.4026)	0.8830** (0.3758)
treat $\times$ yob = 2014		0.8006 (0.6513)	0.6818 (0.6000)	0.6248 (0.5656)
treat $\times$ yob = 2015		0.5968 (0.8029)	0.3803 (0.8051)	0.3005 (0.7810)
treat $\times$ yob = 2016		0.9684** (0.3983)	0.6753** (0.3389)	0.5804* (0.3175)
treat $\times$ yob = 2017		2.558*** (0.5683)	2.484*** (0.5576)	2.437*** (0.5518)
treat $\times$ yob = 2018		3.407*** (0.4524)	3.567*** (0.4620)	3.493*** (0.4911)
treat $\times$ yob = 2019		1.995** (0.8613)	2.083** (0.8848)	2.035** (0.8804)
p-value for nonzero pre-effect		0.772	0.799	0.414
Observations	15,352	15,352	15,352	15,352
R ²	0.16918	0.47434	0.50014	0.50506
Within R ²	0.00575	0.01144	0.04123	0.05067
treat fixed effects	✓	✓	✓	✓
post fixed effects	✓			
yob fixed effects		✓	✓	✓
parent_company fixed effects			✓	✓